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Steensgaard-Madsen

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(45) **Date of Patent: May 3, 2016**

(54) **ANALOG-TO-DIGITAL CONVERTER**

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(21) Appl. No.: **14/162,572**

(22) Filed: **Jan. 23, 2014**

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Related U.S. Application Data

(62) Division of application No. 13/553,092, filed on Jul. 19, 2012, now Pat. No. 8,810,443.

(60) Provisional application No. 61/636,372, filed on Apr. 20, 2012.

(51) **Int. Cl.**

H03M 1/20	(2006.01)
H03M 1/06	(2006.01)
H03M 1/46	(2006.01)

(52) **U.S. Cl.**

CPC ***H03M 1/201*** (2013.01); ***H03M 1/0668*** (2013.01); *H03M 1/0641* (2013.01); *H03M 1/462* (2013.01); *H03M 1/468* (2013.01)

(58) **Field of Classification Search**

CPC H03M 1/0641; H03M 1/12; H03M 3/328; H03M 3/332; H03M 1/0639; H03M 1/0673; H03M 1/806; H03M 1/46; H03M 1/804; H03M 1/1057

USPC 341/118, 120, 131, 144, 143, 155, 172

See application file for complete search history.

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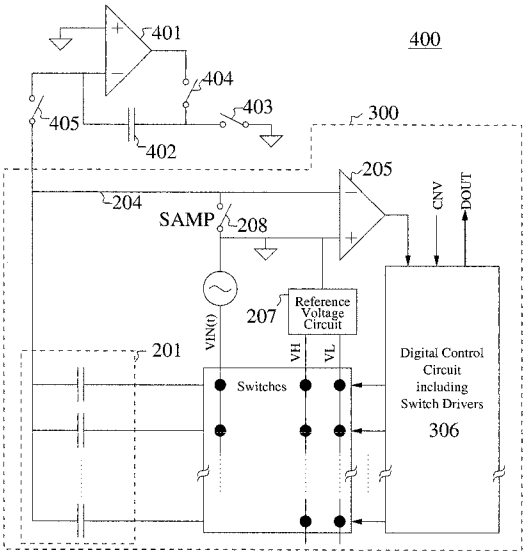
Primary Examiner — Linh Nguyen

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(57) **ABSTRACT**

An analog-to-digital converter (ADC) system and method. The ADC system in accord with one embodiment includes a sampling digital-to-analog converter configured to sample a combination of an analog signal value and an analog dither value, and a control circuit comprising a mismatch-shaping encoder. The control circuit is configured to sequentially apply a plurality of digital codes to the sampling digital-to-analog converter during an analog-to-digital conversion operation to derive a digital code representing the combination of the analog signal value and the analog dither value. Several embodiments are presented.

25 Claims, 11 Drawing Sheets



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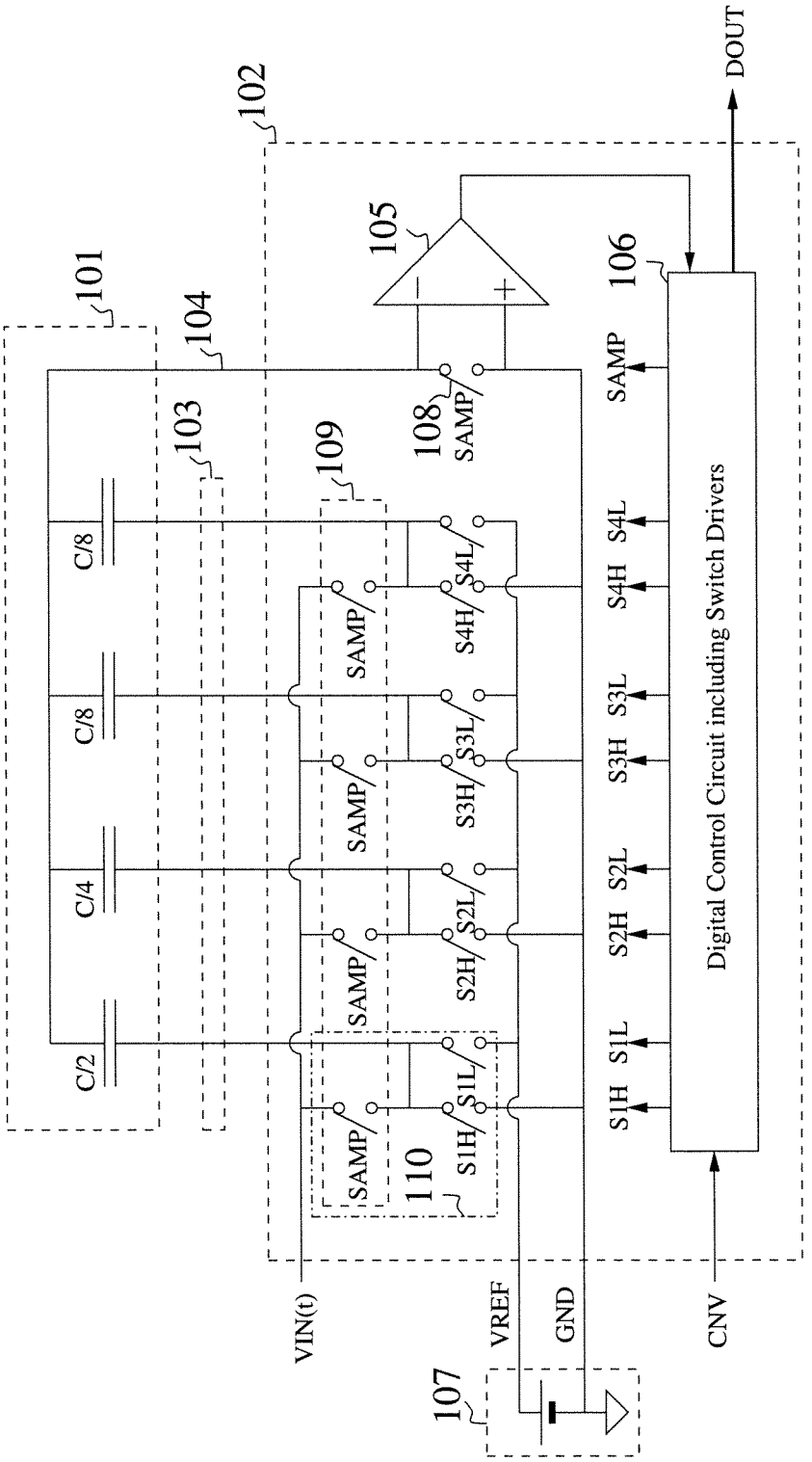


FIGURE 1 (Prior Art)

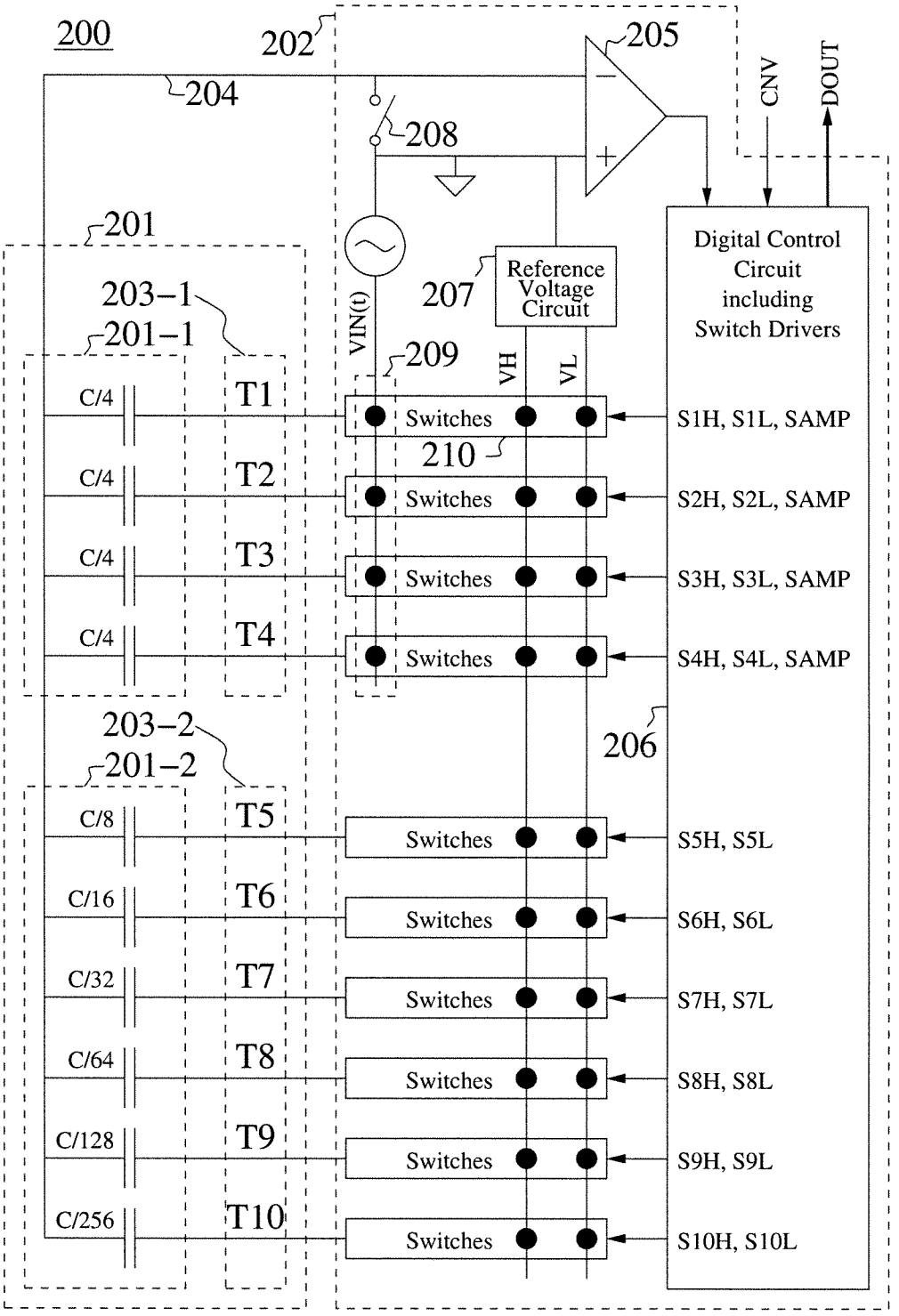

200
202
204
205
208
CNV
DOUT
201
203–1
201–1
207
Reference Voltage Circuit
Digital Control Circuit including Switch Drivers
VIN(t)
209
VH
VL
C/4
T1
Switches
S1H, S1L, SAMP
210
C/4
T2
Switches
S2H, S2L, SAMP
C/4
T3
Switches
S3H, S3L, SAMP
C/4
T4
Switches
S4H, S4L, SAMP
206
203–2
201–2
C/8
T5
Switches
S5H, S5L
C/16
T6
Switches
S6H, S6L
C/32
T7
Switches
S7H, S7L
C/64
T8
Switches
S8H, S8L
C/128
T9
Switches
S9H, S9L
C/256
T10
Switches
S10H, S10L

FIGURE 2

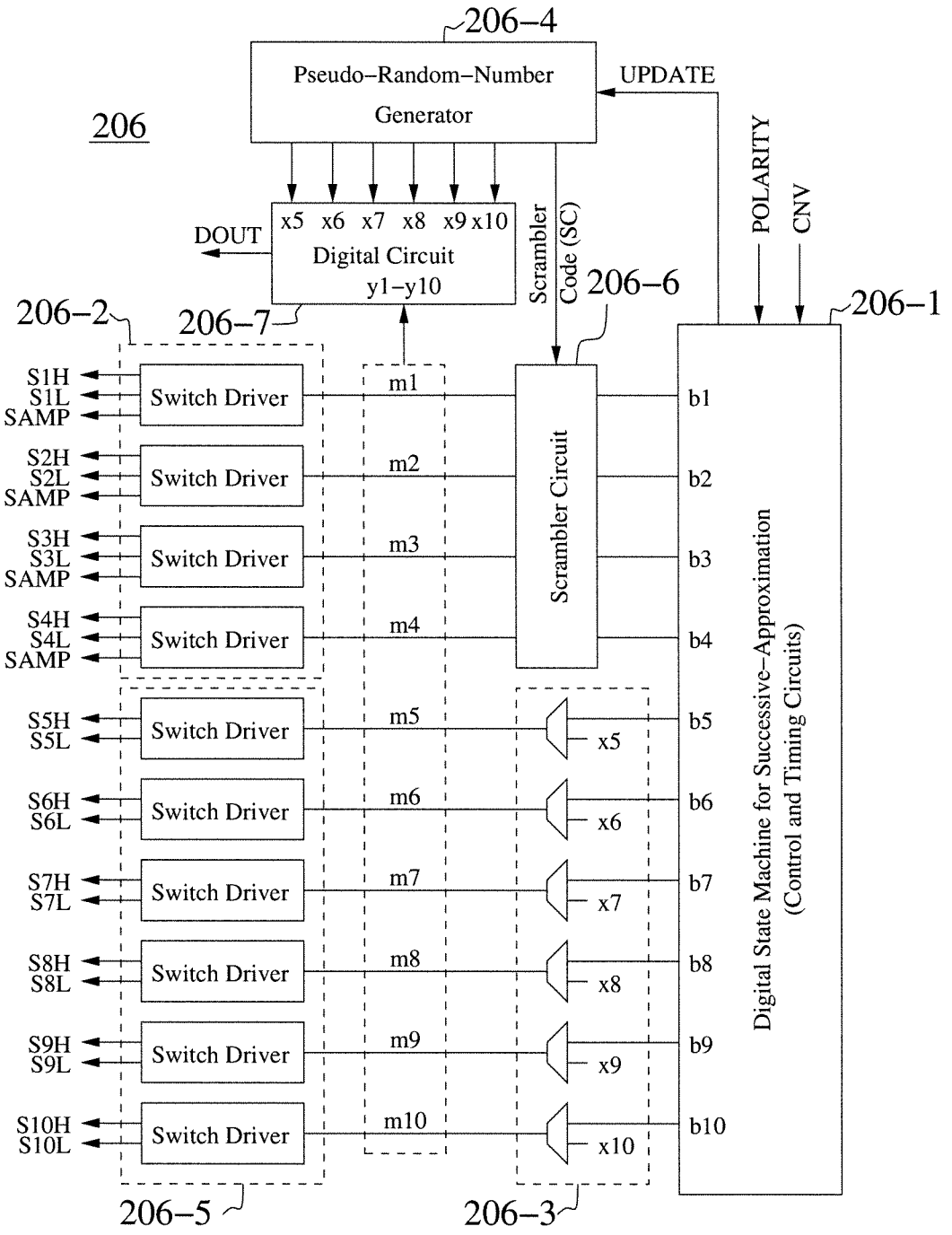

206
206–4
Pseudo–Random–Number Generator
UPDATE
POLARITY
CNV
DOUT
x5 x6 x7 x8 x9 x10
Digital Circuit
y1–y10
Scrambler Code (SC)
206–6
206–1
206–2
206–7
S1H
S1L
SAMP
Switch Driver
m1
b1
S2H
S2L
SAMP
Switch Driver
m2
b2
S3H
S3L
SAMP
Switch Driver
m3
b3
S4H
S4L
SAMP
Switch Driver
m4
b4
Scrambler Circuit
S5H
S5L
Switch Driver
m5
x5
b5
S6H
S6L
Switch Driver
m6
x6
b6
S7H
S7L
Switch Driver
m7
x7
b7
S8H
S8L
Switch Driver
m8
x8
b8
S9H
S9L
Switch Driver
m9
x9
b9
S10H
S10L
Switch Driver
m10
x10
b10
Digital State Machine for Successive–Approximation (Control and Timing Circuits)
206–5
206–3

FIGURE 3

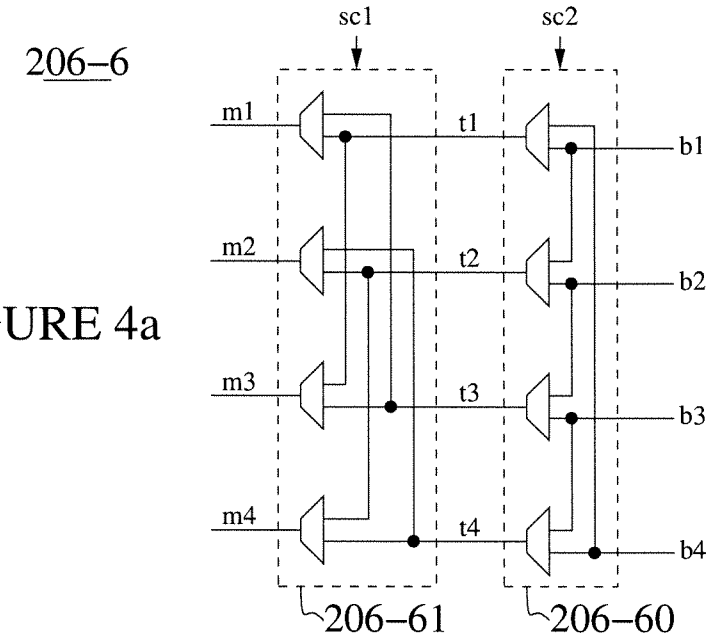


FIGURE 4a

Rotary Scrambler	sc1=0 sc2=0	sc1=0 sc2=1	sc1=1 sc2=0	sc1=1 sc2=1
m1	b1	b4	b3	b2
m2	b2	b1	b4	b3
m3	b3	b2	b1	b4
m4	b4	b3	b2	b1
Sequence	S0	S1	S2	S3

FIGURE 4b

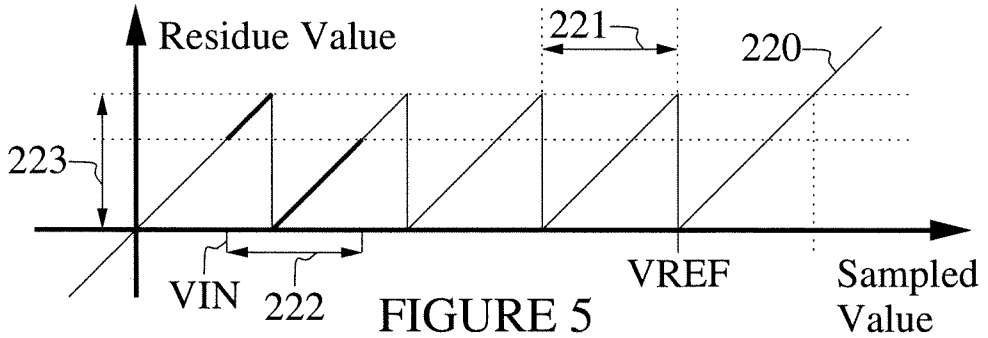


FIGURE 5

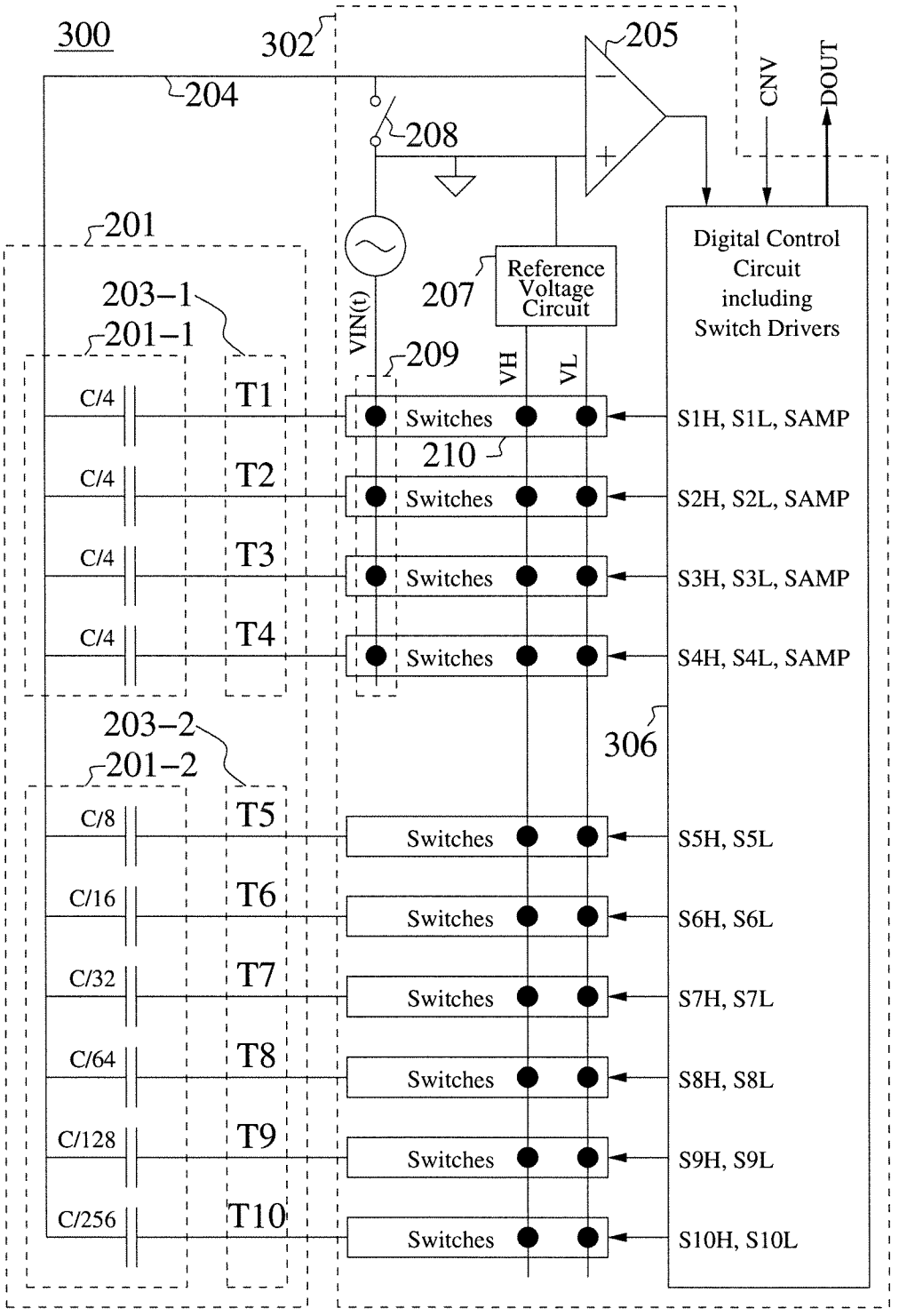

300
302
204
205
208
CNV
DOUT
201
203–1
201–1
207
Reference Voltage Circuit
Digital Control Circuit including Switch Drivers
VIN(t)
209
VH
VL
C/4
T1
Switches
S1H, S1L, SAMP
210
C/4
T2
Switches
S2H, S2L, SAMP
C/4
T3
Switches
S3H, S3L, SAMP
C/4
T4
Switches
S4H, S4L, SAMP
306
203–2
201–2
C/8
T5
Switches
S5H, S5L
C/16
T6
Switches
S6H, S6L
C/32
T7
Switches
S7H, S7L
C/64
T8
Switches
S8H, S8L
C/128
T9
Switches
S9H, S9L
C/256
T10
Switches
S10H, S10L

FIGURE 6

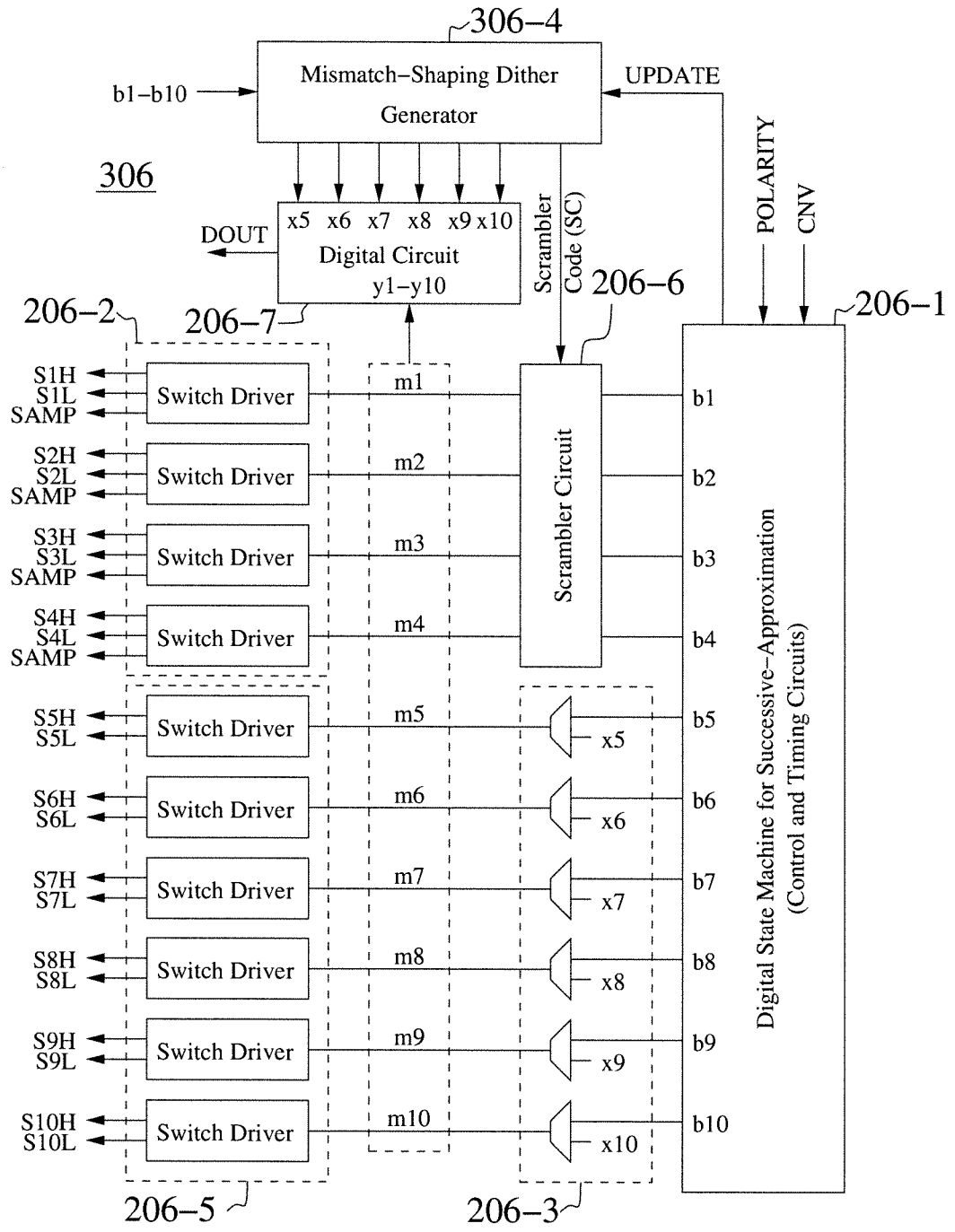

306-4
Mismatch-Shaping Dither Generator
b1-b10
UPDATE
306
DOUT
x5 x6 x7 x8 x9 x10
Digital Circuit
y1-y10
Scrambler Code (SC)
POLARITY
CNV
206-6
206-1
206-2
206-7
S1H
S1L
SAMP
Switch Driver
m1
Scrambler Circuit
b1
S2H
S2L
SAMP
Switch Driver
m2
b2
S3H
S3L
SAMP
Switch Driver
m3
b3
S4H
S4L
SAMP
Switch Driver
m4
b4
S5H
S5L
Switch Driver
m5
x5
b5
S6H
S6L
Switch Driver
m6
x6
b6
S7H
S7L
Switch Driver
m7
x7
b7
S8H
S8L
Switch Driver
m8
x8
b8
S9H
S9L
Switch Driver
m9
x9
b9
S10H
S10L
Switch Driver
m10
x10
b10
Digital State Machine for Successive-Approximation (Control and Timing Circuits)
206-5
206-3

FIGURE 7

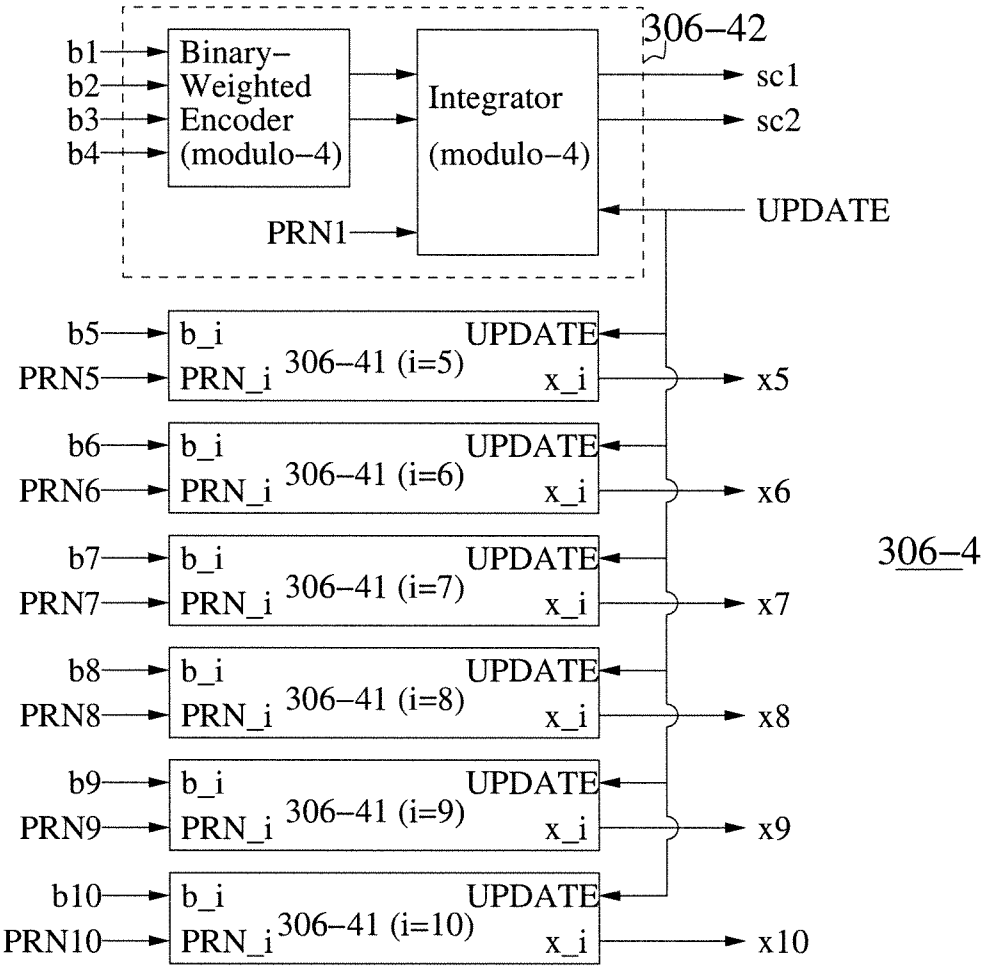


FIGURE 8a

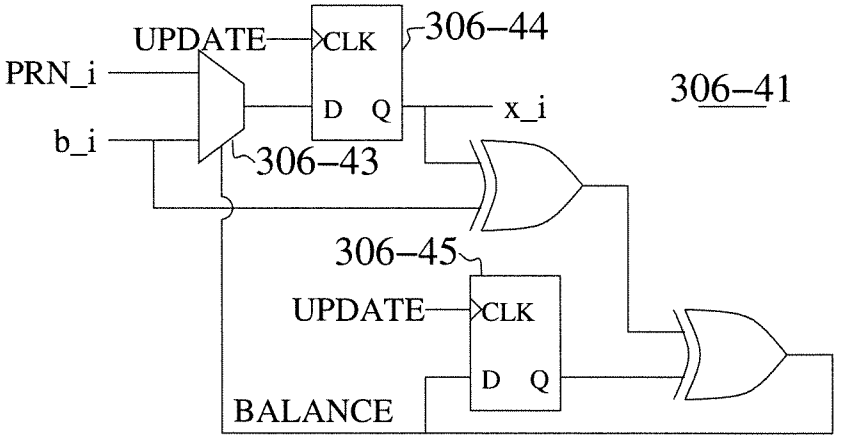


FIGURE 8b

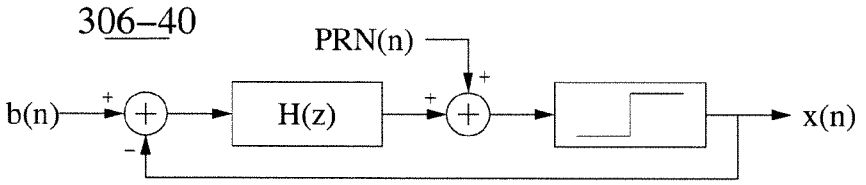


FIGURE 9

FIGURE 10a: Prior–Art SAR ADC

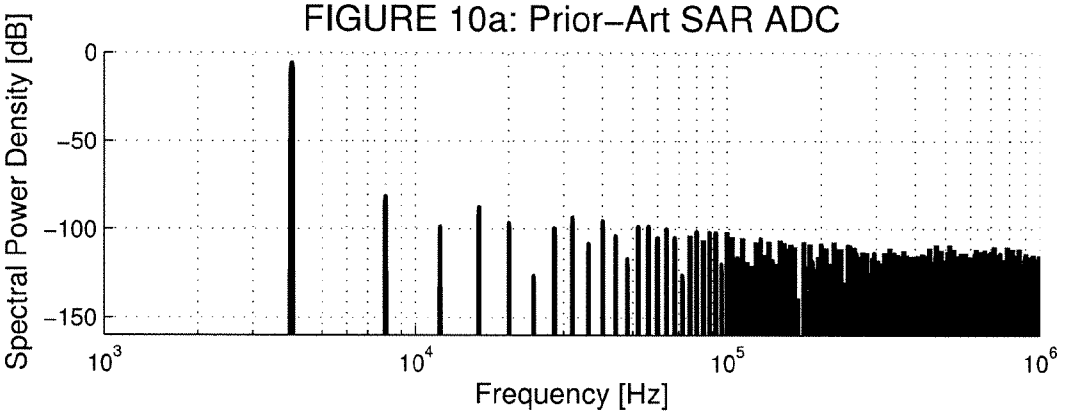


FIGURE 10b: ADC 200 (increased resolution CDAC)

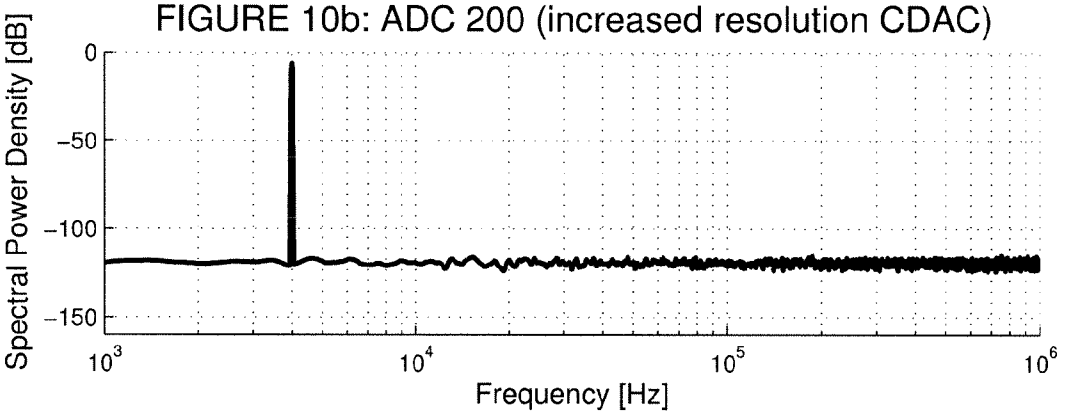
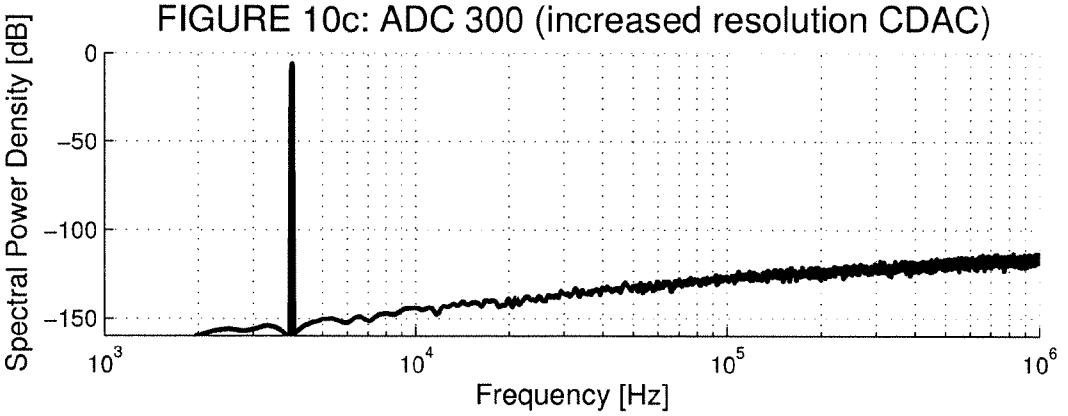


FIGURE 10c: ADC 300 (increased resolution CDAC)



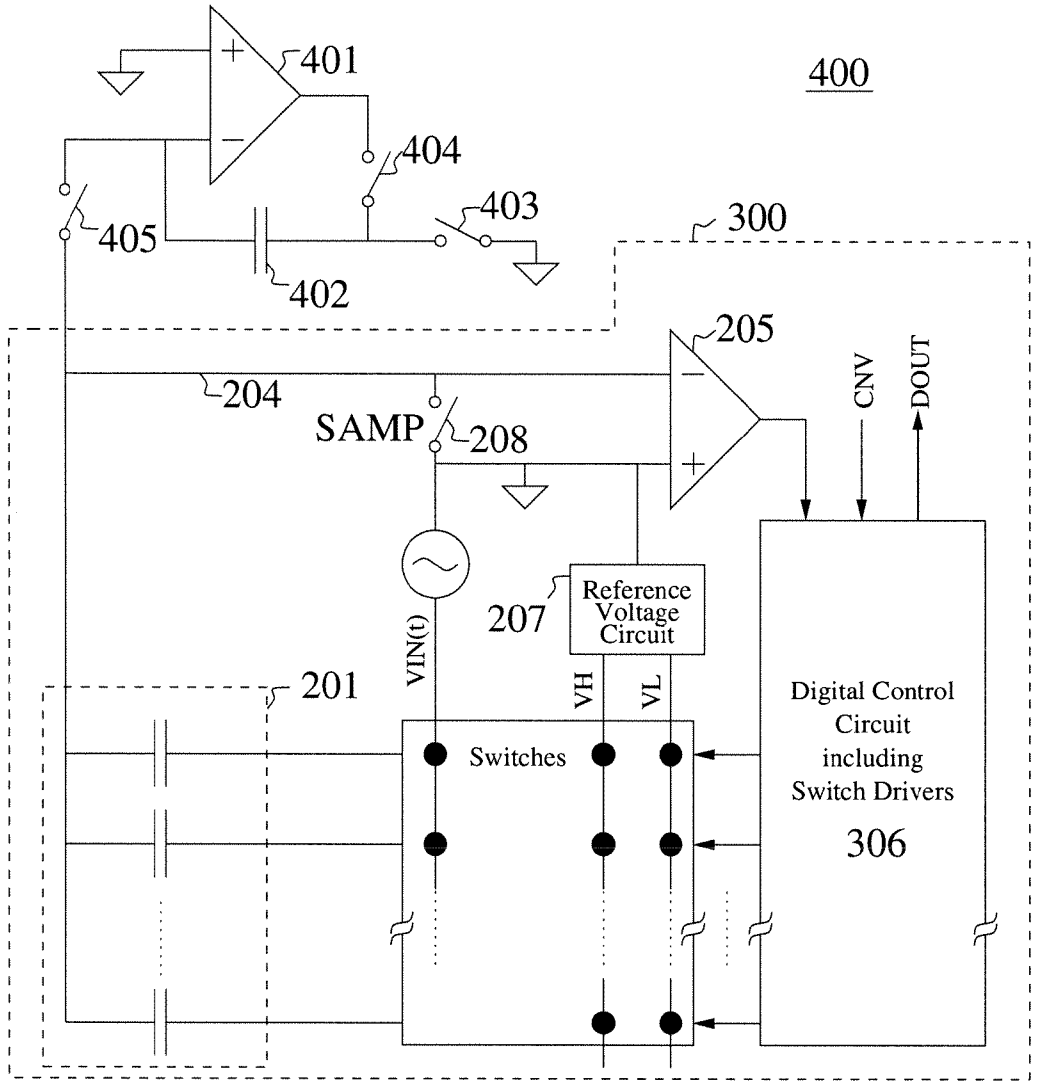


FIGURE 11a

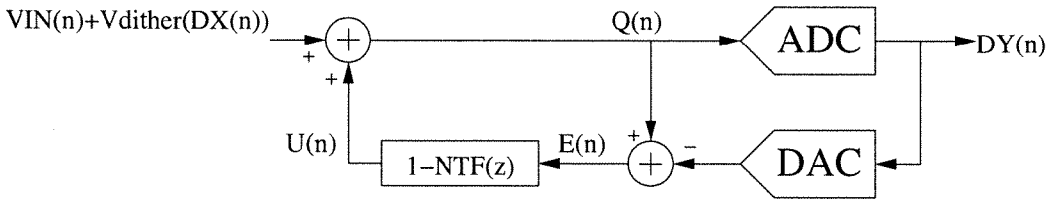


FIGURE 11b

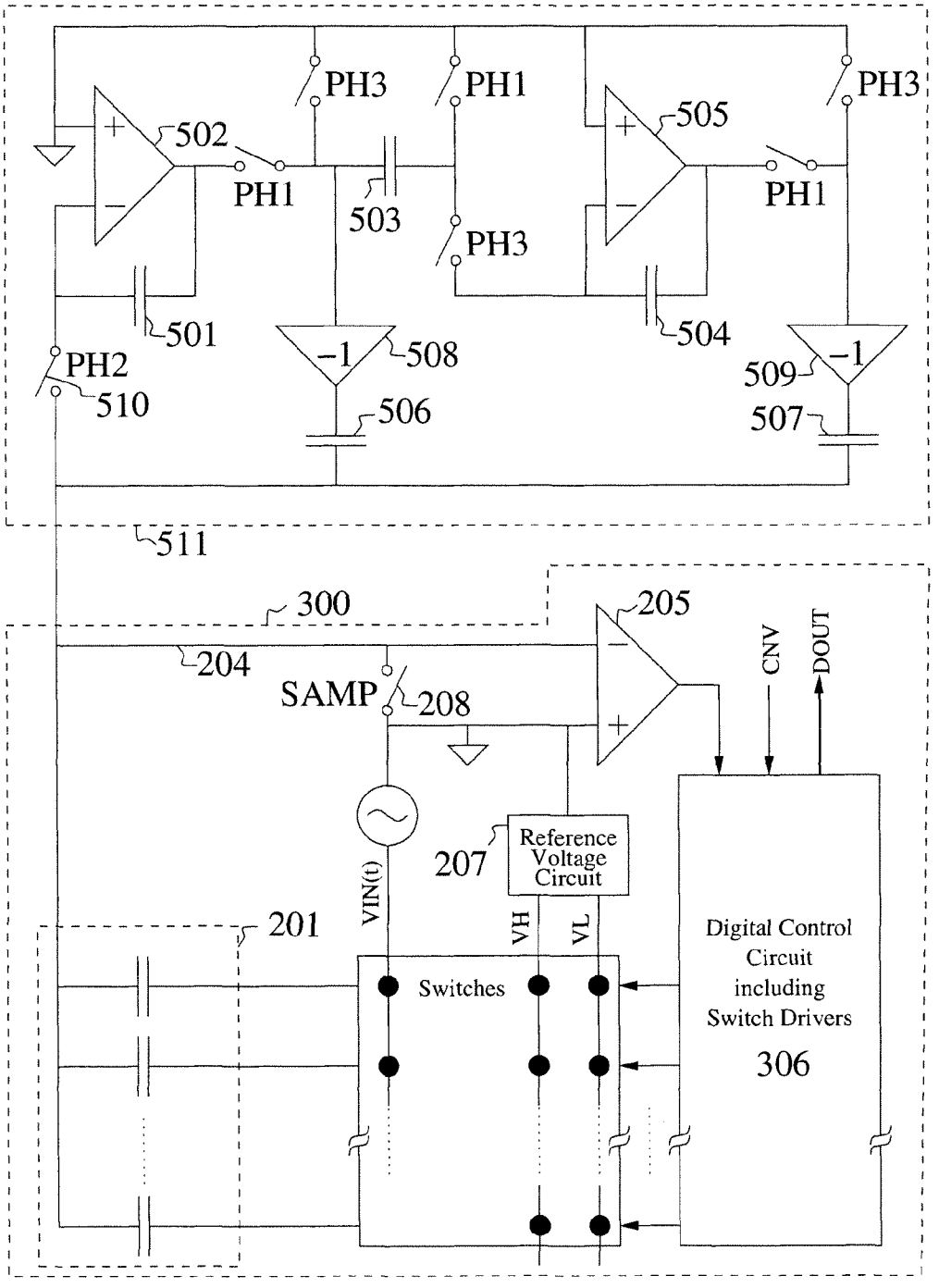

PH3
PH1
PH3
502
505
PH1
PH1
503
PH3
501
504
PH2
510
-1
508
506
509
-1
507
511
300
205
204
SAMP
208
CNV
DOUT
207
Reference Voltage Circuit
VIN(t)
VH
VL
201
Switches
Digital Control Circuit including Switch Drivers
306
500

FIGURE 12a

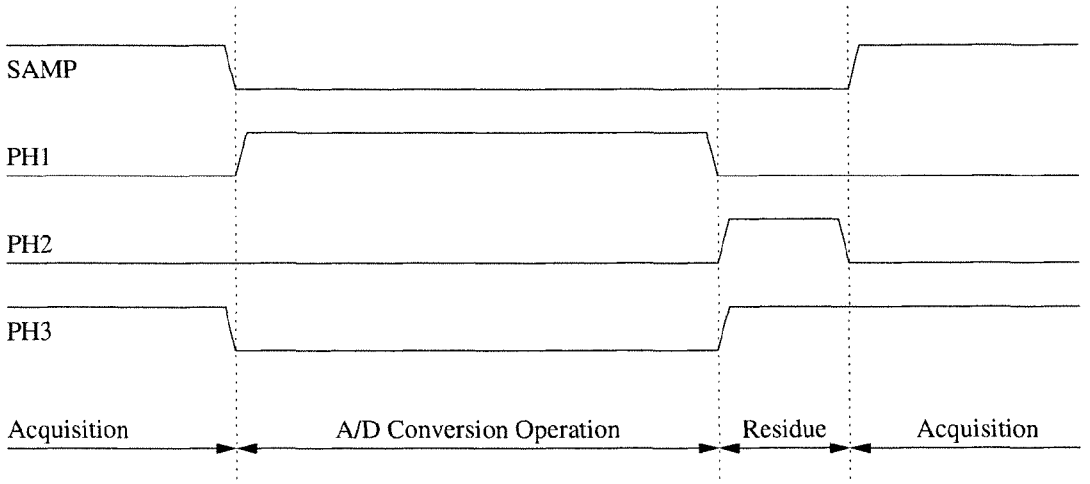


FIGURE 12b

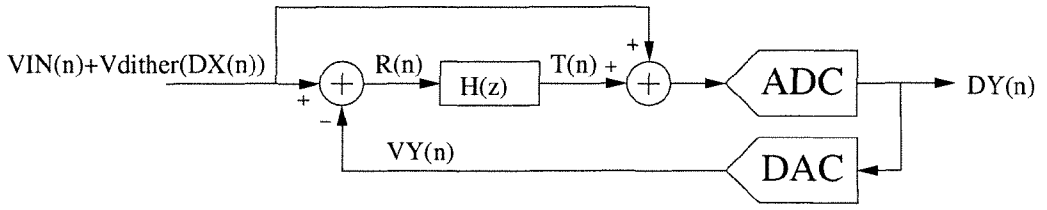


FIGURE 12c

ANALOG-TO-DIGITAL CONVERTER

RELATED APPLICATION

This Application is a Divisional Application of U.S. Ser. No. 13/553,092, filed Jul. 19, 2012, which claims priority from U.S. Provisional Application No. 61/636,372 filed Apr. 20, 2012, the disclosure of each Application incorporated by reference herein.

BACKGROUND

1. Technical Field

The present teaching relates to analog circuits and methods. Particularly, the present teaching relates to analog-to-digital converters (ADCs) and operating thereof.

2. Discussion of Technical Background

Analog-to-digital converters (ADCs) are used for a wide range of applications, including, but not limited to, sensor interfaces, industrial applications, consumer applications, and communications. Various circuits and techniques have been developed for analog-to-digital (A/D) conversion targeting various applications and their varying requirements in terms of speed, resolution, noise, power consumption, and other performance related parameters.

FIG. **1** shows a prior-art successive-approximation-register (SAR) analog-to-digital converter (SAR ADC) **100**. Successive approximation is a well-known sequential method used for A/D conversion, where an analog signal value VIN may be sampled on a capacitive digital-to-analog converter (CDAC) **101**, and a sequential successive-approximation A/D conversion operation is used to generate an encoded numerical (digital) representation DOUT of the analog signal value VIN. A control circuit **102** may apply a sequence of digital codes to a plurality of input terminals **103** of CDAC **101** causing an output **104** of CDAC **101** to converge towards a predefined value (e.g., GND=0V). A comparator circuit **105** may provide an indication of a polarity of CDAC output **104**. A digital control circuit **106** may use the indication of polarity to select a next digital code in the sequence of digital codes applied to CDAC input terminals **103**. CDAC output **104** may represent a residue of analog signal value VIN with respect to a digital code applied to CDAC **101** incorporating a reference voltage VREF. The reference voltage may be embedded in a physical representation of a digital code. For example, a high state (logic "one") of a bit of a digital code may be represented physically by a first reference voltage potential VH=VREF applied to an input terminal **103**; likewise, a low state (logic "zero") may be represented by applying a second reference voltage potential VL=GND=0V to the input terminal **103**. Accordingly, a high/low logic value of a bit (a bit value) of a digital code may be applied to an input terminal of a CDAC by circuitry such as logic gates, switch drivers, and switches connecting the input terminal to a reference voltage circuit **107** providing a plurality of reference voltage potentials. When a residue is made to converge towards zero, a final digital code in a sequence of digital codes may be a digital representation of an analog signal value VIN. A digital circuit may combine bit values of the final digital code to provide an encoded numerical representation DOUT, which may be provided in a standardized format using standard logic levels (e.g., serial communication at 1.8V CMOS logic levels of a binary-weighted code).

A sampling node **104** of a CDAC (which is also an output **104** of CDAC **101** in FIG. **1**, but a sampling node may be distinct from an output) may be coupled to a predefined potential (say ground, GND=0V) by a sampling switch **108**

during an acquisition period. A charge portion may be substantially isolated on sampling node **104** at a sampling instance when sampling switch **108** is opened (i.e., when sampling switch **108** is controlled to be substantially non-conductive). By coupling an analog voltage signal VIN(t) to at least one CDAC input terminal **103** during an acquisition period, a value VIN of voltage signal VIN(t) at a sampling instance is effectively sampled on CDAC **101** (a substantially isolated charge portion is a sampled value representing VIN). Sampling switch **108** and one-or-more input switches **109** may be controlled by digital control circuit **106**. An applied logic signal CNV may be used to control when to sample analog voltage signal VIN(t) and perform an A/D conversion of a sampled value VIN.

A selectable scaling of a sampled representation of an analog signal value VIN may be achieved by applying analog voltage signal VIN(t) to a selectable subset of CDAC input terminals **103** during an acquisition period. This aspect is described in U.S. Pat. No. 8,130,133, which is incorporated herein by reference for describing such scaling and the structure and operation of a prior-art successive-approximation ADC.

An input terminal of a CDAC may be assigned a weighting factor for characterizing how much a voltage variation at the input terminal affects a charge portion that may be isolated at a sampling node at a sampling instance. Weighting factors may alternately be viewed as (scaled) voltage gain factors from CDAC input terminals to a CDAC output when a sampling switch is open. For example, exemplary CDAC **101** of FIG. **1** may have four input terminals **103** with weighting factors (from left to right) w1=0.5, w2=0.25, w3=0.125, and w4=0.125. If an input voltage VIN=1.25V is applied to all four input terminals **103** at a sampling instance, a charge portion on sampling node **104** may be (−Qsamp)=C*VIN*(w1+w2+w3+w4)=C*VIN. For a 5V reference voltage (first reference potential VREF=5V; second reference potential GND=0V) an identical value for Qsamp corresponds to a digital code '0100', which may alternately be described by 4 bit values b1=0, b2=1, b3=0, and b4=0. Accordingly, applying digital code '0100' to exemplary CDAC **101** may cause sampling node **104** to resume a predefined potential applied to it at a sampling instance when VIN=1.25V. This property reflects that VIN/VREF=(b1*w1+b2*w2+b3*w3+b4*w4)/(w1+w2+w3+w4)=0.25. A digital code may be said to correspond to an analog signal value VIN (for a given CDAC and reference voltage), or alternately that the digital code is a representation of analog signal value VIN (the digital representation based on a reference voltage and a plurality of weighting factors).

A quantization error may depend on a resolution of a CDAC. A digital code may be said to represent an analog signal value VIN when a residue on sampling node **104** is within a predefined range corresponding to a resolution. Accordingly, a digital code may correspond to (represent) any signal value in a range, not just a specific value for which a quantization error is exactly zero. A CDAC may have a relatively high resolution (say 20 bits), and a digital code may thus correspond to any signal value in a relatively narrow range.

A numerical value representing VIN/VREF may be calculated for a CDAC when it is operated in a manner similar to that described herein, when the CDAC is characterized by a set of weighting factors, and when a digital code corresponding to VIN is known. A plurality of digital codes may be equivalent, in a sense that they all substantially correspond to a single analog signal value VIN. For example, digital code '0011' is equivalent to digital code '0100' with respect to

exemplary CDAC **101** of FIG. **1**; both digital codes represent VIN/VREF=0.25. Nominally, it does not matter which of the two codes is provided by control circuit **102** for an A/D conversion operation of VIN=1.25V.

A CDAC for which several digital codes are equivalent may provide a measure of redundancy for an A/D conversion operation. Redundancy may facilitate a control circuit to recover from certain errors that may occur during a successive-approximation ADC operation, without having to go back to a step in the operation at which such errors may have occurred. For example, over-ranging techniques exemplify a use of redundancy to facilitate an ADC to recover from some errors. Over-ranging techniques are well known and may be used in pipeline ADCs, SAR ADCs, and many other types of ADCs. Several CDAC structures that facilitate over-ranging during an A/D conversion operation are described in US Patent Application Publication US2011/0115661 A1, which is incorporated herein by reference for describing over-ranging techniques, for describing several CDAC structures, and for describing ADC circuits that may be used in combination with the present teaching.

Persons skilled in the art can analyze a CDAC structure, identify its weighting factors, and devise a method to derive a suitable encoding of a numerical value represented by a digital code corresponding to an analog signal value with respect to the CDAC. A suitable encoding may use binary-weighted digital codes to represent numerical values. Many other suitable encoding techniques are known to persons skilled in the art. An encoding technique for reducing a latency parameter is described in US Patent Application Publication US 2011/0285567 A1, which is incorporated herein by reference for describing encoding techniques and for describing over-ranging techniques. Likewise, many structures for CDACs are known in the art, including CDACs utilizing capacitive voltage division to realize very small weighting factors, and also including CDACs wherein a resistive DAC circuit structure may be used to provide scaled voltages at nodes that are capacitively coupled to a sampling node. Accordingly, a capacitive digital-to-analog converter (CDAC) shall refer to any digital-to-analog converter structure (whether or not purely capacitive) having a sampling node at which a charge portion can be substantially isolated at a sampling instance, and having a plurality of input terminals that may be substantially characterized by weighting factors. Accordingly, an analog signal value can be sampled on a CDAC, and the CDAC may be substantially characterized by weighting factors assigned to each input terminal in a plurality of input terminals. These properties characterize a general class of circuits, sampling digital-to-analog converters, which include CDACs.

An accuracy of a numerical value representing VIN/VREF derived from a digital code applied to a CDAC characterized by a set of (assumed) weighting factors depends on how accurately the assumed weighting factors used for a calculation of the numerical value represent a set of actual weighting factors of the physical CDAC structure. A difference between the assumed weighting factors and the actual weighting factors may be referred to as "mismatch of weighting factors" or "weighting factor mismatch". Mismatch of weighting factors may (for example) be caused by mismatch of capacitor ratios in a CDAC. If assumed weighting factors are determined at design time, before a CDAC circuit is manufactured, weighting factor mismatch may be affected by manufacturing repeatability. Imperfect manufacturing repeatability of a CDAC may substantially degrade overall accuracy of an analog-to-digital converter system. A lower degree of weighting factor mismatch, and thus better overall accuracy, may be achieved by estimating (measuring) actual weighting factors of a CDAC after it has been manufactured. U.S. Pat. No. 7,705,765 describes how weighting factors of a CDAC may be measured, and how digital codes representing measured weighting factors may be stored and combined with a digital code from a A/D conversion operation to derive an encoded numerical value representing an analog signal value. U.S. Pat. No. 7,705,765 is incorporated herein by reference for describing how to measure, store, and apply codes representing weighting factors of a CDAC used in an analog-to-digital converter system, for describing over-ranging techniques, and for describing several CDAC structures and the implementation and operation of an ADC system based on successive approximation.

The circuits and methods described in U.S. Pat. No. 7,705,765 may facilitate very good estimation of CDAC weighting factors, and highly accurate ADC systems incorporating accurately estimated (assumed) weighting factors may be realized. However, a high degree of overall accuracy can be ensured only for as long as the actual weighting factors do not change substantially after the estimation process. Environmental changes (such as temperature variations) and other effects may cause some degree of weighting factor mismatch that may degrade an overall accuracy of an ADC for which weighting factors are not measured frequently. It may not be practical or desirable to interrupt an operation of an analog-to-digital converter system to measure weighting factors sufficiently frequently to ensure a long-term accurate operation.

What is needed is an analog-to-digital converter system that has a reduced sensitivity to weighting factor mismatch of a CDAC, such that highly accurate analog-to-digital converters can be implemented effectively and operated without interruption.

SUMMARY

In accord with one embodiment of the presented teachings, an analog-to-digital converter system comprises a sampling digital-to-analog converter configured to sample a combination of an analog signal value and an analog dither value, and a control circuit comprising a mismatch-shaping encoder. The control circuit is configured to sequentially apply a plurality of digital codes to the sampling digital-to-analog converter during an analog-to-digital conversion operation to derive a digital code.

In accord with another embodiment, an analog-to-digital converter system comprises a sampling digital-to-analog converter configured to sample a first analog value comprising a portion of an analog signal value and a portion of an analog dither value; a dither generator circuit configured to apply a digital dither code to the sampling digital-to-analog converter at a sampling instance, and a control circuit configured to derive a first digital code to represent the first analog value, the control circuit is further configured to combine the first digital code and the digital dither code to derive an encoded numerical representation of the analog signal value. The dither generator circuit is configured to be responsive to a preceding digital code derived during a preceding analog-to-digital conversion operation to represent an analog value.

In accord with a further embodiment of an analog-to-digital converter (ADC), the ADC comprises a capacitive digital-to-analog converter configured to sample a combination of an analog signal value and an analog dither value, and a control circuit which comprises a scrambler circuit for scrambling bit values. The control circuit sequentially applies a plurality of digital codes to the capacitive digital-to-analog converter dur-

ing a conversion operation to derive an encoded numerical representation of the analog signal value.

In accord with yet a further embodiment, an analog-to-digital converter system comprises a sampling digital-to-analog converter configured to sample an analog value and to provide a representation of a residue of the sampled analog value with respect to a digital code, a plurality of registers configured to store codes representing weighting factors of the sampling digital-to-analog converter measured to account for variations in a manufacturing process, and a control circuit comprising a mismatch-shaping encoder. The control circuit is configured to derive and apply a first digital code to the sampling digital-to-analog converter during an analog-to-digital conversion operation. Further included is a digital circuit configured to combine bit values of the first digital code with codes stored in the plurality of registers to derive a digital output code.

In accord with another embodiment, an analog-to-digital converter operating in a sampling phase and a conversion phase comprises a sampling digital-to-analog converter having a first plurality of input terminals and a second plurality of input terminals; and a control circuit comprising a mismatch-shaping encoder. In the sampling phase, the control circuit is configured to couple an analog input signal to at least one terminal in the first plurality of input terminals and to apply a digital dither code to the second plurality of input terminals. In the conversion phase, the control circuit is configured to sequentially apply a plurality of digital codes to the sampling digital-to-analog converter to derive an encoded numerical representation of a value of the analog input signal at a sampling instance.

In accord with another embodiment, a capacitive digital-to-analog converter is configured to sample a first analog value comprising a portion of an analog signal value and a portion of an analog dither value. A dither generator circuit applies a digital dither code to the capacitive digital-to-analog converter during an acquisition period. A control circuit is configured to derive a first digital code to represent the first analog value. The control circuit combines the first digital code and the digital dither code to derive an encoded numerical representation of the analog signal value. The dither generator circuit is responsive to a preceding digital code derived during a preceding analog-to-digital conversion operation to represent an analog value.

A method for deriving an encoded numerical representation of an analog signal value using a sampling digital-to-analog converter having a plurality of input terminals, in accord with an embodiment of this disclosure, comprises isolating a first charge portion on a sampling node of the sampling digital-to-analog converter while applying a first digital dither code to at least one input terminal in the plurality of input terminals; applying a first digital code to the sampling digital-to-analog converter for which a potential of the sampling node is within a predefined range; generating a second digital dither code responsive to the first digital code; and combining the first digital dither code and the first digital code to derive the encoded numerical representation of the analog signal value.

BRIEF DESCRIPTION OF THE DRAWINGS

The present teachings claimed and/or described herein are further described in terms of exemplary embodiments. These exemplary embodiments are described in detail with reference to drawings. These embodiments are non-limiting exemplary embodiments, in which like reference numerals represent similar structures throughout the several views of the drawings, and wherein:

FIG. **1** (Prior Art) illustrates a successive-approximation analog-to-digital converter (ADC) **100**;

FIG. **2** illustrates a first preferred embodiment of the present teaching ADC **200**;

FIG. **3** illustrates a digital control circuit **206** incorporated in ADC **200** (FIG. **2**);

FIG. **4***a* illustrates a scrambler circuit **206-6** incorporated in digital control circuit **206** (FIG. **3**);

FIG. **4***b* illustrates a truth table for permutations (which may be described as sequences) provided by scrambler circuit **206-6** (FIG. **4***a*) versus scrambler codes;

FIG. **5** illustrates a nominal saw-tooth relationship between a sampled value and a residue after bit values b1, b2, b3, b4 are determined in ADC **200** (FIG. **2**);

FIG. **6** illustrates a second preferred embodiment of the present teaching ADC **300**;

FIG. **7** illustrates a digital control circuit **306** incorporated in ADC **300** (FIG. **6**);

FIG. **8***a* illustrates a mismatch-shaping dither generator **306-4** incorporated in digital control circuit **306** (FIG. **7**);

FIG. **8***b* illustrates a mismatch-shaping dither generator **306-41** incorporated in mismatch-shaping dither generator **306-4** (FIG. **8***a*);

FIG. **9** illustrates a block diagram of a mismatch-shaping dither generator **306-40** that may be incorporated in a mismatch-shaping dither generator **306-4** (FIG. **8***a*);

FIG. **10***a* illustrates a performance of a prior-art successive-approximation SAR ADC;

FIG. **10***b* illustrates a performance of an embodiment of the present teachings, ADC **200** with an increased resolution CDAC;

FIG. **10***c* illustrates a performance of an embodiment of the present teachings, ADC **300** with an increased resolution CDAC;

FIG. **11***a* illustrates a third preferred embodiment of the present teachings ADC **400**;

FIG. **11***b* illustrates a signal-flow model for ADC **400** (FIG. **11***a*).

FIG. **12***a* illustrates a fourth preferred embodiment of the present teachings ADC **500**;

FIG. **12***b* illustrates a timing diagram for ADC **500** (FIG. **12***a*);

FIG. **12***c* illustrates a signal-flow model for ADC **500** (FIG. **12***a*).

DETAILED DESCRIPTION

FIG. **2** shows a first preferred embodiment of an analog-to-digital converter (ADC) **200** according to the present teachings. A structure of ADC **200** has several similarities to prior-art ADC **100** of FIG. **1**. The structure of ADC **200** will be described first with reference to ADC **100**. Several important differences between ADCs **100** and **200** will be described subsequently.

ADC **200** comprises a CDAC **201** and a control circuit **202** (compare to CDAC **101** and control circuit **102** of ADC **100**). CDAC **201** comprises a first plurality of capacitors **201-1** and a second plurality of capacitors **201-2**, which are connected by a sampling node **204** on which a charge portion may be substantially isolated when a sampling switch **208** is opened at a sampling instance (compare to sampling node **104** and sampling switch **108** of ADC **100**). First plurality of capacitors **201-1** is coupled to control circuit **202** via a first plurality of input terminals **203-1**. Second plurality of capacitors **201-2**

is coupled to control circuit **202** via a second plurality of input terminals **203-2** (compare to input terminals **103** and control circuit **102** of ADC **100**). A comparator circuit **205** is configured to indicate a polarity of a potential of sampling node **204**. A digital control circuit **206** is configured to receive the indication of polarity and to provide a sequence of digital codes applied to CDAC **201** via CDAC input terminals **203-1**, **203-2** (compare to comparator circuit **105**, digital control circuit **106**, and input terminals **103** of ADC **100**). Digital codes applied to CDAC **201** are represented physically by well-controlled reference voltage potentials (VH=VREF represents logic "one", VL=GND=0V represents logic "zero") provided by a reference voltage circuit **207** (compare to CDAC **101** and reference voltage circuit **107** of ADC **100**). Digital control circuit **206** is configured to drive switches to couple input terminals **203-1**, **203-2** to either an input voltage signal VIN(t) or a reference voltage potential provided by reference voltage circuit **207**. A group of switches **210** in ADC **200** is equivalent to a group of switches **110** in ADC **100**. Switches **210** in FIG. **2** are represented by a graphical representation that is more compact than a graphical representation of switches **110** in FIG. **1**. A more compact representation of switches **210** is used in FIG. **2** to better illustrate important aspects (details) of the present teaching. It will be understood by persons skilled in the art that a group of selectable connections **209** in FIG. **2** illustrate a structure and a function equivalent to that of input switches **109** in FIG. **1**.

A change in an applied logic signal CNV initiates an A/D conversion operation, starting with sampling a combination of an analog signal value VIN and an analog dither value on CDAC **201**. Digital control circuit **206** is configured to be responsive to CNV and is configured to open sampling switch **208** at a sampling instance to substantially isolate a charge portion on sampling node **204**. A substantially isolated charge portion on node **204** is a sampled value representing a combination of a plurality of voltage potentials applied to input terminals **203-1**, **203-2** at the sampling instance. Digital control circuit **206** is configured to control switches to couple input voltage signal VIN(t) to input terminals T1, T2, T3, and T4 in the first plurality of input terminals **203-1** during an acquisition period preceding the sampling instance. Digital control circuit **206** is further configured to control switches to apply a substantially random digital code to input terminals T5, T6, T7, T8, T9, and T10 in the second plurality of input terminals **203-2** during the acquisition period.

A pseudo-random-number (PRN) generator is included in digital control circuit **206** and configured to produce a plurality of substantially random coin-toss bit value sequences. A conventional linear-feedback-shift-register (LFSR) and/or a cellular-automata-shift-register (CASR) circuit may be used to produce a sequence of sufficiently random bit values (a long periodic LFSR/CASR sequence may be characterized as "substantially random" or "sufficiently random"). Digital control circuit **206** is configured to apply a substantially random 6-bit code DX to CDAC input terminals **203-2** via switch control signals (S5H, S5L, S6H, S6L, S7H, S7L, S8H, S8L, S9H, S9L, S10H, S10L) during an acquisition period. DX may be described by 6 bit values x5, x6, x7, x8, x9, x10, each having an index referring to an index of a corresponding CDAC input terminal. For example, if DX='001011' (x5=0, x6=0, x7=1, x8=0, x9=1, x10=1) switches are configured to couple reference voltage potential VH to input terminals T7, T9, and T10 and reference voltage potential VL to input terminals T5, T6, T8. CDAC **201** may be characterized by a plurality of weighting factors characterizing how substantially random code DX corresponds to a substantially random analog "dither" value Vdither(DX) sampled on CDAC **201** at a sampling instance.

A charge portion that is sampled on CDAC **201** (substantially isolated on sampling node **204** at a sampling instance) is a sampled value representing a combination of an analog signal value VIN applied to CDAC **201** via terminals **203-1** and a substantially random analog dither value Vdither(DX) applied to CDAC **201** via terminals **203-2**. Accordingly, CDAC **201** is configured to sample a combination of an analog signal value and an analog dither value.

CDAC **201** may be characterized by a plurality of nominal weighting factors w(T1)=¼, w(T2)=¼, w(T3)=¼, w(T4)=¼ , w(T5)=⅛, w(T6)=1/16, w(T7)=1/32, w(T8)=1/64, w(T9)=1/128, w(T10)=1/256. Weighting factors may be scaled by any factor for convenience (without changing a physical structure CDAC **201**). It may be convenient to scale weighting factors such that a sum of a selected group of weighting factors is unity. A scaling is selected such that w(T1)+w(T2)+w(T3)+w(T4)=1.

Control circuit **202** is configured to apply a sequence of digital codes to CDAC **201** to derive a 10-bit digital code DY corresponding to a combined signal and dither value sampled on CDAC **201** at a sampling instance. Digital code DY may be characterized by 10 bit values y1, y2, y3, . . . , y10 each having an index referring to an index of a corresponding CDAC input terminal. Control circuit **202** is configured to calculate DOUT as a weighted sum of bit values and bit-value-differences DOUT=y1*w(T1)+y2*w(T2)+y3*w(T3)+y4*w(T4)+(y5−x5)*w(T5)+(y6−x6)*w(T6)+(y7−x7)*w(T7)+(y8−x8)*w(T8)+(y9−x9)*w(T9)+(y10−x10)*w(T10). For example, for DX='100011' and DY='1011_011010', control circuit **202** is configured to calculate DOUT=(1+0+1+1)/4+(0−1)/8+(1−0)/16+(1−0)/32+(0−0)/64+(1−1)/128+(0−1)/256=183/256.

CDAC **201** is redundant and weighting factors w(T1), w(T2), w(T3), w(T4) are nominally identical. A permutation of bit values y1, y2, y3, y4 for any code DY does not change a nominal value of DOUT. Accordingly, an ordering of bit values y1, y2, y3, y4 does not affect DOUT or a nominal operation of ADC **200**. For example, DY codes '1100_000000', '0110_000000' '0011_000000', '1010_000000' are nominally equivalent and represent DOUT=0.5 for DX='000000'.

Digital control circuit **206** is configured to use a successive-approximation search algorithm to derive a code DY representing a value sampled on CDAC **201**. Bit values y1, y2, y3, y4, are determined in one of several distinct sequences to derive one of several nominally equivalent codes. A first sequence S0=[y1,y2,y3,y4] will be used to describe an operation. In a first step (sequence S0), a first digital code '1000_000000' is applied to CDAC **201** and bit value y1 is set according to a polarity indicated by comparator circuit **205**. For example, if a negative polarity of sampling node **204** is indicated, then y1 is set to 1; otherwise y1 is set to 0. In a second step, a second digital code 'y1_100_000000' is applied and bit value y2 is set according to a polarity indicated by comparator circuit **205**. In a third step, a third digital code 'y1_y2_10_000000' is applied and bit value y3 is set according to a polarity indicated by comparator circuit **205**. The successive-approximation A/D conversion process continues in this manner for 10 steps to determine 10 bits values, resulting in a code DY='y1_y2_y3_y4_y5_y6_y7_y8_y9_y10' representing an analog value sampled on CDAC **201**.

A sequence for bit values y1, y2, y3, y4 is selected on a substantially random basis by bit values provided by a PRN generator. A plurality of predefined sequences is supported by a hardware implementation of digital control circuit **206**. N

items can be ordered in N-factorial distinct sequences, and hence up to 4!=24 distinct sequences may be considered for identifying y1, y2, y3, y4 using a successive approximation algorithm. It may be advantageous to select a lower number of sequences to be supported by a digital control circuit. Digital control circuit **206** supports **4** distinct sequences: S0=[y1, y2, y3, y4]; S1=[y2, y3, y4, y1]; S2=[y3, y4, y1, y2]; S3=[y4, y1, y2, y3]. For each new A/D conversion operation, one of the 4 distinct sequences is selected on a substantially random basis to be used for the A/D conversion operation. For example, assume sequence S2 is selected by a state of a PRN generator. In a first step of a conversion operation for sequence S2, digital control circuit **206** is configured to apply a first digital code '0010_000000' to CDAC **201** and set bit value y3 according to a polarity indicated by comparator circuit **205**. In a second step of a conversion operation for sequence S2, digital control circuit **206** is configured to apply a second digital code '00_y3_1_000000' to CDAC **201** and set bit value y4 according to a polarity indicated by comparator circuit **205**. In a third step of a conversion operation for sequence S2, digital control circuit **206** is configured to apply a third digital code '10_y3_y4_000000' to CDAC **201** and set bit value y1 according to a polarity indicated by comparator circuit **205**. In a fourth step of a conversion operation for sequence S2, digital control circuit **206** is configured to apply a fourth digital code 'y1_1_y3_y4_000000' to CDAC **201** and set bit value y2 according to a polarity indicated by comparator circuit **205**. The successive-approximation A/D conversion operation continues with six additional steps, to determine bit values y5, y6, y7, y8, y9, y10 in sequence.

FIG. **3** shows a structure of digital control circuit **206**. During an acquisition period, a digital state machine for successive approximation **206-1** is configured to control a first plurality of switch drivers **206-2** (a control signal is not shown explicitly) to provide switch control signals SAMP, S1H, S1L, S2H, S2L, S3H, S3L, S4H, S4L controlling switches, including sampling switch **208** and input switches **209**, to apply analog voltage signal VIN(t) across first plurality of capacitors **201-1** (see FIG. **2**). Digital state machine **206-1** is further configured to control a plurality of digital multiplexer circuits **206-3** (a control signal is not shown explicitly), such that a plurality of substantially random bit values x5, x6, x7, x8, x9, x10 provided by a PRN generator **206-4** is coupled to a second plurality of switch drivers **206-5** providing switch control signals S5H, S5L, S6H, S6L, S7H, S7L, S8H, S8L, S9H, S9L, S10H, S10L controlling switches coupled to the second plurality to input terminals **203-2** of CDAC **201** (see FIG. **2**). Digital state machine **206-1** is configured to be responsive to an applied logic signal CNV, which may transition to mark a sampling instance and initiate an A/D conversion operation. In response to a change in CNV, digital state machine **206-1** controls switch drivers **206-2** to open sampling switch **208** and input switches **209** to isolate a charge portion on sampling node **204** (see FIG. **2**). Switch drivers **206-2**, **206-5** are then controlled to provide switch control signals to apply a digital code M composed of a plurality of bit values m1, m2, m3, m4, m5, m6, m7, m8, m9, m10 to CDAC **201**. Digital multiplexer circuit **206-3** is configured to set bit values m5, m6, m7, m8, m9, m10 equal to a plurality of bit values b5, b6, b7, b8, b9, b10 provided by digital state machine **206-1**. A scrambler circuit **206-6** is configured to receive a scrambler code SC from PRN generator **206-4**, and to provide bit values m1, m2, m3, m4 as a permutation of a plurality of bit values b1, b2, b3, b4 provided by digital state machine **206-1**. Scrambler code SC selects m1, m2, m3, m4 as a predefined permutation of b1, b2, b3, b4 corresponding to a predefined sequence for y1, y2, y3, y4. For example, sequence S2=[y3, y4, y1, y2] may be selected by configuring scrambler circuit **206-6** such that m3=b1; m4=b2; m1=b3; m2=b4. Digital state machine **206-1** is configured to implement a successive-approximation-register search algorithm (testing b1=1 in a first step; testing b2=1 in a second step; and so on) after sampling a value in response to a change in CNV.

Consider a conversion operation for which scrambler code SC is set to select sequence S2. In a first step, m3=b1=1 and a first code M='0010_000000' is applied to CDAC **201**. The first step is complete when y3=m3=b1 is set according to a polarity indicated by comparator circuit **205**. In a second step, m4=b2=1 and a second code M='00_y3_1_000000' is applied to CDAC **201**. The second step is complete when y4=m4=b2 is set according to a polarity indicated by comparator circuit **205**. Digital state machine **206-1** continues the successive-approximation algorithm for each remaining bit in sequence b3, b4, b5, b6, b7, b8, b9, b10. When each bit value b1-b10 has been determined, bit values y1=m1, y2=m2, y3=m3, y10=m10 (collectively DY=M) are combined with bit values x5, x6, x7, x8, x9, x10 (collectively DX) as described earlier. A digital circuit **206-7** comprises latches to capture bit values of digital codes DX and DY at an end of a conversion. Digital circuit **206-7** further comprises a combinatorial circuit for calculating DOUT=y1*w(T1)+y2*w(T2)+y3*w(T3)+y4*w(T4)+(y5−x5)*w(T5)+(y6−x6)*w(T6)+(y7−x7)*w(T7)+(y8−x8)*w(T8)+(y9−x9)*w(T9)+(y10−x10)*w(T10). Digital circuit **206-7** may be configured with memory circuits (e.g., registers) to store digital codes for weighting factors of CDAC **201**, which may be estimated after ADC **200** is manufactured to account for variations in a manufacturing process. Digital state machine **206-1** then controls PRN generator **206-4** to update/renew scrambler code SC and bit values of DX for a next acquisition period and A/D conversion operation.

FIG. **4***a* shows scrambler circuit **206-6** in more detail. Scrambler code SC comprises a first bit value sc1 and a second bit value sc2. A plurality of bit values t1, t2, t3, t4 are a first permutation (controlled by sc2) of bit values b1, b2, b3, b4. A first digital multiplexer circuit **206-60** is configured to provide t1=b1, t2=b2, t3=b3, t4=b4 for sc2=0 and t2=b1, t3=b2, t4=b3, t1=b4 for sc2=1. A second digital multiplexer circuit **206-61** is configured to provide bit values m1, m2, m3, m4 as a permutation of bit values t1, t2, t3, t4 according to sc1. Specifically, digital multiplexer circuit **206-61** is configured such that m1=t1, m2=t2, m3=t3, m4=t4 for sc1=0 and m3=t1, m4=t2, m1=t3, m2=t4 for sc1=1. FIG. **4***b* shows a mapping of bit values b1, b2, b3, b4 to bit values m1, m2, m3, m4 for all scrambler codes. It may be recognized that sc2 controls a 1-position-rotation of a code 'b1_b2_b3_b4' and that sc1 controls a 2-position-rotation of a code 't1_t2_t3_t4'. Accordingly, it may be recognized that a numerical value N of SC='sc1_sc2' controls a N-position-rotation of code 'b1_b2_b3_b4'. A scrambler circuit that substantially rotates an input code to derive a permutation output code may be referred to as a "rotary scrambler".

Scrambler circuit **206-6** implements a function known in the art as "dynamic element matching", which was described by L. Richard Carley in IEEE Journal of Solid-State Circuits, Volume 24, April 1989. An aspect of dynamic element matching is that errors caused by mismatch of weighting factors w(T1), w(T2), w(T3), w(T4) (which may be referred to as mismatch-induced errors) are scrambled and may be sufficiently randomized to substantially assume noise-like characteristics. Statistical properties of a sequence of substantially random scrambler codes SC and a configuration of scrambler circuit **206-6** may be selected such that a bit value

b1 (in any given A/D conversion operation) is substantially equally likely to be coupled to any one of terminals T1, T2, T3, T4. Accordingly, an effective weighting factor of a representation of a sequence b1(n) (the representation included in a sequence of residues on sampling node **204**) may be substantially a mean value of w(T1), w(T2), w(T3), and w(T4). Similar properties may characterize b1(n), b2(n), b3(n), and b4(n), each of which may be substantially represented by the same effective mean-value weighting factor.

A sequence of numerical representations DOUT(n) of a constant analog input signal VIN(t) may comprise a sequence of mismatch-induced errors ERR(n) caused by mismatch of weighting factors w(T1), w(T2), w(T3), w(T4). For a conventional SAR ADC **100** (FIG. **1**), a sequence of mismatch-induced errors may be constant ERR(1)=ERR(2)=ERR(3)=. . . , and an average of several values DOUT(n) may not substantially improve an accuracy of a numerical representation. For ADC **200** (FIGS. **2** and **3**), scrambler circuit **206-6** is configured to induce a sequence of mismatch-induced errors to vary from one sample to another, and an average of several values DOUT(n) may be a numerical representation of VIN having an improved accuracy.

A sequence of mismatch-induced errors ERR(n) may be of a substantially stochastic nature, and may be described by parameters commonly used to describe stochastic signals, such as noise. An objective of dynamic element matching may be to reduce a correlation of an analog signal VIN(t) and a sequence of mismatch-induced errors ERR(n) from A/D converting the analog signal VIN(t). L. Richard Carley described in 1989 that a mismatch-induced error sequence may be (substantially) a white-noise sequence with zero mean and having a variance depending on a degree of mismatch of weighting factors. Accordingly, an improved accuracy may be achieved by an averaging operation applied to a sequence of digital codes DOUT(n), even if the sequence DOUT(n) is from converting an input signal VIN(t) that may not be constant. An averaging operation (including a non-uniform weighted averaging operation) may be implemented as a digital filter processing DOUT(n). A degree to which accuracy is improved may depend on a degree to which a digital filter is configured to limit a noise bandwidth. A digital filter for reducing a noise bandwidth may be a finite-impulse-response (FIR) or an infinite-impulse-response (IIR) type digital filter effectively applying a sequence of impulse-response weighting coefficients to DOUT(n). Multi-rate digital filtering techniques may also be used.

If weighting factor mismatch is reduced to a low level (e.g., by estimating and applying weighting factors as described in U.S. Pat. No. 7,705,765) a variance of a mismatch-induced noise-like error sequence may be less than a variance of a thermal (or other) noise process characterizing an ADC circuit. Accordingly, it is optional to apply an averaging operation to improve accuracy. An embodiment of the present teachings may be configured to scramble a mismatch-induced error sequence to induce white-noise-like properties of such errors, and overall accuracy may be substantially determined by another white-noise-like process (e.g. thermal noise).

The dynamic element matching technique and circuit described by L. Richard Carley in 1989 has since been generalized to a class of dynamic element matching techniques and circuits known as "mismatch-shaping encoders". Scrambler circuit **206-6** (FIGS. **3**, **4***a* and **4***b*) and the circuits described by L. Richard Carley may be classified as zero-order mismatch-shaping encoders, which nominally randomize mismatch-induced errors into a substantially white-noise-like error signal. First, second, and higher-order mismatch-shaping encoders nominally randomize mismatch-induced errors into a noise-like error signal having a substantially non-uniform spectral power density (e.g., having a relatively lower spectral power density in a signal band). First-order mismatch-shaping encoders configured to randomize mismatch-induced errors into a noise-like error signal having a relatively lower spectral power density at low frequencies are described in U.S. Pat. No. 6,348,884, which is included herein by reference for describing mismatch-shaping encoders. Scrambler circuit **206-6** (FIG. **3**) may be substituted by another scrambler circuit for a mismatch-shaping encoder of any order. A mismatch-shaping encoder may be configured to randomize mismatch-induced errors into a noise-like signal having a relatively lower spectral power density at frequencies that are within a pass-band (signal-band) of a digital filter processing a sequence of digital codes DOUT(n). A pass-band may include a base-band range of frequencies (e.g., 0 Hz to 100 kHz), or it may be a band-pass range of frequencies (e.g., from 100 kHz to 200 kHz).

Scrambler circuit **206-6** is configured to substantially randomize mismatch-induced errors caused by mismatch of weighting factors for terminals T1, T2, T3, T4. An effective mean-value weighting factor may characterize a plurality of terminals coupled to a scrambler circuit.

Mismatch of weighting factors for the second plurality of input terminals **203-2** (T5, T6, T7, T8, T9, T10) with respect to an effective mean-value weighting factor characterizing the first plurality of input terminals **203-1** may also cause mismatch-induced errors. An objective of sampling an analog dither value on CDAC **201** by coupling substantially random bit values x5, x6, x7, x8, x9, x10 to input terminals **203-2** during acquisition periods may be to reduce a correlation of an analog signal VIN(t) with sequences of bit values y5(n), y6(n), y7(n), y8(n), y9(n), y10(n). For example, if a bit-value-difference sequence y5(n)−x5(n) is a white-noise-like sequence, then w(T5)*(y5(n)−x5(n)) may be a white-noise-like sequence for any fixed weighting factor w(T5), including a mismatched weighting factor. Accordingly, mismatch-induced errors from mismatch of weighting factors for input terminals **203-2** may be suppressed by an averaging operation limiting a noise bandwidth.

FIG. **5** shows a nominal functional relationship **220** of a residue value (vertical axis) for a value sampled on CDAC **201** (horizontal axis) after bit values b1, b2, b3, and b4 are determined by digital state machine **206-1** in ADC **200** (FIGS. **2** and **3**). The sampled value may be expressed as (w(T1)+w(T2)+w(T3)+w(T4))*VIN+(w(T5)*x5+w(T6)*x6+w(T7)*x7+w(T8)*x8+w(T9)*x9+w(T10)*x10)*VREF or more compactly (for a selected scaling of weighting factors) as VIN+Vdither(DX). VIN is a value of VIN(t) sampled on sampling node **204** of CDAC **201** in combination with an analog dither value Vdither(DX). VIN may be any value in a full-scale range from 0V to VREF. An exemplary value VIN is marked on the horizontal axis in FIG. **5**. If each bit value x5, x6, x7, x8, x9, x10 of digital dither code DX is a substantially random coin-toss sequence, then analog dither value Vdither(DX)=(w(T5)*x5+w(T6)*x6+w(T7)*x7+w(T8)*x8+w(T9)*x9+w(T10)*x10)*VREF may be a substantially stochastic quantity that may be substantially uniformly distributed in a range that is approximately as wide as a tooth-spacing **221** (VREF/4) of saw-tooth characteristic **220** in FIG. **5**. Accordingly, a combination of VIN and an analog dither value sampled on node **204** may be substantially uniformly distributed in a range **222** marked on the horizontal axis. In approximation, a residue value corresponding to a stochastic variable uniformly distributed in range **222** may be a stochastic variable distributed in a range **223**. Range **222** is mapped to range **223** by a high-lighted (by a relatively thicker

line width) section of saw-tooth characteristic **220**. Range **223** and stochastic properties of a residue value may be substantially independent of a value VIN in a range from 0V to VREF. At an end of an A/D conversion process, bit values y5, y6, y7, y8, y9, y10 may represent a residue value, and each bit value may be from a substantially stochastic coin-toss sequence. Accordingly, stochastic properties of an analog dither value Vdither(DX) may substantially randomize and induce stochastic properties for bit values y5, y6, y7, y8, y9, y10. A pseudo-stochastic variable DX may be provided by pseudo-random-number generator **206-4** to provide a substantially uniformly distributed pseudo-stochastic analog dither value Vdither(DX) that may be combined with VIN and sampled on CDAC **201** to induce (in approximation) desired pseudo-stochastic properties for bit values y5, y6, y7, y8, y9, y10. Accordingly, bit-value-difference sequences y5(n)−x5(n), y6(n)−x6(n), y10(n)−x10(n), may be substantially described as stochastic variables, and may be (in approximation) white-noise sequences.

Properties of Vdither(DX) may be selected in various ways to induce various desired statistical properties for mismatch-induced errors. For example, Vdither(DX) may be uniformly distributed in a range that is approximately as wide as one or more teeth of saw-tooth characteristic **220**, or Vdither(DX) may be triangularly distributed in a range that is two teeth wide. Vdither(DX) may, but need not, be an uncorrelated (white-noise) stochastic sequence. Next, an embodiment will be described wherein Vdither(DX) is generated and applied to randomize and shape (a spectral power density of) mismatch-induced errors.

FIG. **6** shows a second embodiment of the present teachings ADC **300**. An aspect of ADC **300** is that a mismatch-induced error sequence that may be caused by mismatch of weighting factors w(T1), w(T2), w(T10) may be induced to be a substantially stochastic noise-like error signal having a substantially non-uniform spectral power density. A total power of a mismatch-induced error sequence may be related to a degree of mismatch of weighting factors. A non-uniform spectral power density may provide that a relatively small fraction of a total power of a mismatch-induced error signal may be present in a signal band of ADC **300**. A digital filter (not shown) may be configured to suppress spectral power of DOUT(n) outside the signal band. The digital filter may thus provide an improved accuracy by limiting a noise bandwidth. ADC **300** may be configured to calculate DOUT(n) based on a plurality of assumed weighting factors determined at design time (before ADC **300** is manufactured). A noise-like mismatch-induced error signal having a substantial total power may be suppressed to below a thermal-noise level by limiting a noise bandwidth. In another embodiment, weighting factors may be estimated and stored after ADC **300** is manufactured to reduce a total power of a mismatch-induced error signal. Accordingly, a spectral power density of a mismatch-induced error signal may be lower than a spectral power density of a thermal noise in a signal band only, or it may be lower than a thermal noise at all frequencies in a Nyquist range of frequencies. It may be advantageous to use ADC **300** in either case.

ADC **300** of FIG. **6** is substantially identical to ADC **200** of FIG. **2**, with the exception that digital control circuit **206** is substituted by another digital control circuit **306**. Digital control circuit **306** shown in FIG. **7** is substantially identical to digital control circuit **206** of FIG. **3**, with the exception that pseudo-random-number generator **206-4** in **206** is substituted by a mismatch-shaping dither generator **306-4** in **306**. Similar to PRN generator **206-4**, mismatch-shaping dither generator **306-4** is configured to provide sequences of scrambler codes SC and digital dither codes DX coupled to scrambler circuit **206-6** and digital multiplexer circuit **206-3**. A difference between ADC **200** of FIG. **2** and ADC **300** of FIG. **6** relates to properties imposed on DX (bit values x5, x6, x7, x8, x9, x10) and SC (bit values sc1, sc2), to induce desired approximate statistical properties on mismatch-induced errors.

Mismatch-shaping dither generator **306-4** is configured to receive bit values b1, b2, b10 provided by digital state machine **206-1** at an end of an A/D conversion operation. Bit values b1, b2, b3, b4 may be a thermometer-type code 'b1_b2_b3_b4' (any one of codes '0000', '1000', '1100', '1110', '1111'). A thermometer-type code 'b1_b2_b3_b4' may be rotated according to scrambler code SC such that scrambler circuit **206-6** implements a first-order mismatch-shaping operation described in U.S. Pat. Nos. 5,221,926 or 6,348,884. Any other mismatch-shaping encoder circuit may be considered for substituting scrambler circuit **206-6**. Some mismatch-shaping encoders may generate scrambler codes internally (i.e., they may be configured to include a portion of **306-4**, and some mismatch-shaping encoders may be configured to receive an input signal encoded using binary-weighted codes (e.g., U.S. Pat. No. 5,684,482). A theory, implementation and operation of such mismatch-shaping encoders is well known by those skilled in the art, and need not be described in further detail.

Mismatch-shaping dither generator **306-4** is configured to provide digital dither codes DX such that an error that may be caused by mismatch of weighting factors of input terminals **203-2** is induced to be a substantially stochastic error sequence characterized by a spectral power density that is relatively lower in a selected signal band.

Digital circuit **206-7** is configured to calculate encoded numerical values DOUT=y1*w(T1)+y2*w(T2)+y3*w(T3)+y4*w(T4)+(y5−x5)*w(T5)+(y6−x6)*w(T6)+(y7−x7)*w(T7)+(y8−x8)*w(T8)+(y9−x9)*w(T9)+(y10−x10)*w(T10). Spectral components of a mismatch-induced error sequence may include spectral components of bit-value-difference sequences: y5(n)−x5(n), y6(n)−x6(n), . . . , y10(n)−x10(n). Mismatch-shaping dither generator **306-4** is configured to induce desired spectral properties for each bit-value-difference sequence. A polarity of a next value for a bit-value-difference sequence may be selected prior to a sampling instance. Specifically, a bit value (e.g., x5) may be selected to be logic one to induce a non-positive polarity value in a bit-value-difference sequence (e.g., y5−x5) and may be selected to be logic zero to induce a non-negative polarity value in the bit-value-difference sequence. For example, if mismatch-shaping dither generator **306-4** provides x5=0, then y5−x5=y5−0 will be either 0 or 1 (i.e., non-negative) depending on how b5=y5 is set during an A/D conversion operation. If mismatch-shaping dither generator **306-4** provides x5=1, a resulting value of y5−x5=y5−1 will be either (−1) or 0 (i.e., non-positive).

FIG. **8***a* shows mismatch-shaping dither generator **306-4** in more detail. Six instances of a mismatch-shaping dither generator **306-41** (FIG. **8***b*) are configured to generate bit values x5, x6, x7, x8, x9, x10 from bit values b5, b6, b7, b8, b9, b10 and substantially random bit values PRN5, PRN6, PRN7, PRN8, PRN9, PRN10 provided by a pseudo-random-number generator (not shown). A mismatch-shaping scrambler code generator **306-42** is configured to integrate binary-weighted values B(n)=b1(n)+b2(n)+b3(n)+b4(n) modulo-4. Mismatch-shaping scrambler code generator **306-42** may be configured to multiplex B(n) to two integration-modulo-4 circuits to suppress potential idle tones (see U.S. Pat. No. 6,348,884). Mismatch-shaping scrambler code generator **306-42** is configured to provide scrambler codes SC (bit values sc1 and sc2) to scrambler circuit **206-6** shown in FIGS. **4***a*, **4***b*, and **7**.

FIG. **8***b* shows mismatch-shaping dither generator **306-41** in more detail. Mismatch-shaping dither generator **306-41** is configured to generate a bit-value x_i (e.g., x5 for i=5) from b_i (e.g., b5 for i=5) and a random bit value PRN_i (e.g., PRN5 for i=5). A next bit-value of x_i is provided by a flip-flop (FF) circuit **306-44** when UPDATE (see FIGS. **7** and **8***a*) transitions to indicate that an A/D conversion operation is complete and that a valid bit value b_i (e.g., b5=y5 for i=5) is provided as an input to mismatch-shaping dither generator **306-41**. A binary signal BALANCE controls a digital multiplexer circuit **306-43** such that an input signal to FF **306-44** is b_i when BALANCE=0 and PRN_i when BALANCE=1. A second FF **306-45** is also clocked by UPDATE. FF **306-45** is configured to preserve a logic state stored in FF **306-45** when bit values x_i and b_i are identical, and to toggle the logic state when bit values x_i and b_i are different.

Consider an initial state of mismatch-shaping dither generator **306-41** for which BALANCE=1 at an UPDATE event. Input PRN_i of multiplexer **306-43** is coupled to FF **306-44**, causing a random value PRN_i to be stored in FF **306-44** and applied as a next bit value x_i. A logic one (BALANCE=1) is stored in FF **306-45**. If an A/D conversion operation provides b_i=x_i, then BALANCE=1 and the described scenario repeats at a next UPDATE event when a next random value PRN_i is stored in FF **306-44** for a next A/D conversion operation. At some point, an A/D conversion operation may provide a bit value b_i that is not identical to x_i, in which case an UPDATE event causes BALANCE=0 to be stored in FF **306-45** and bit value b_i is stored in FF **306-44** and provided as x_i. Flip-flop circuits **306-44** and **306-45** are configured to retain their logic states until an UPDATE event occurs for which bit values and x_i are again different (causing BALANCE=1 at an UPDATE event, the initial state).

Accordingly, mismatch-shaping dither generator **306-41** may have two modes of operation. In a first mode of operation (BALANCE=0), a bit value x_i of a digital dither code DX is set equal to a bit value of a digital code DY derived in a preceding A/D conversion operation. In a second mode of operation (BALANCE=1), bit value x_i is set on a substantially random basis. Mismatch-shaping dither generator **306-41** is configured to switch between the first and second modes of operation when x_i and b_i are different.

Mismatch-shaping dither generator **306-41** is configured to ensure that a running sum (integral) of a bit-value-difference sequence b_i(n)−x_i(n) is always in a range from negative one to positive one (for a selected origin of time n=0), and mismatch-shaping dither generator **306-41** may thus be characterized as a circuit controlling a spectral component of a bit-value-difference sequence b_i(n)−x_i(n) (at 0 Hz). Mismatch-shaping dither generator **306-41** is configured to provide a substantially random bit value when the running sum of b_i(n)−x_i(n) is zero (a predefined condition characterized by BALANCE=1), and to otherwise provide x_i as b_i delayed (another predefined condition characterized by BALANCE=0). Accordingly, a bit-value-difference sequence provided by mismatch-shaping dither generator **306-41** may be substantially randomized and have a relatively lower spectral power density in a selected signal band.

In another simplified embodiment, mismatch-shaping dither generator **306-41** may be substituted by a single flip-flop providing x_i by delaying a previous value of b_i, In other words, another mismatch-shaping dither generator may be equivalent to mismatch-shaping dither generator **306-41** operating only in the first mode of operation (BALANCE=0). A bit-value-difference sequence may be expressed as b_i(n)−x_i(n)=b_i(n)−b_i(n−1), and thus spectral components may be suppressed in a signal band by a difference operation. For some applications, mismatch-shaping dither generator **306-41** may provide a more effective randomization of the bit-value-difference sequence.

FIG. **9** shows a block diagram for a mismatch-shaping dither generator **306-40**. The block diagram reflects that a mismatch-shaping dither generator **306-40** may be designed explicitly as a closed-loop control system. For a set of design parameters, mismatch-shaping dither generator **306-41** may be modeled as **306-40**, and the circuits are thus substantially equivalent. An input signal b(n) may be a single bit value signal (e.g. b_i in FIG. **8***b*) or a encoded numerical value comprising multiple bit values (e.g., b(n) may be a sequence of codes each comprising 3 bit values 'b5_b6_b7'). Likewise, a digital dither sequence x(n) may be a sequence of single bit values (e.g., x_i in FIG. **8***b*) or it may be a sequence of numerical values represented by codes comprising several bit values. (e.g., 'x5_x6_x7'). A difference between numerical values represented by b(n) and numerical values represented by x(n) is an input to a filter H(z). Filter H(z) may be of any order, and may have a frequency response that may be characterized as low-pass, band-pass, or high-pass. For example, a preferred embodiment may provide a second-order filter H(z)=(1+0.25/(z+1))/(z+1). Filter H(z) may comprise explicit multi-bit registers for storing a binary-weighted encoded state variable, or it may comprise a plurality of flip-flop circuits for storing a less conventional representation of a state variable (e.g., flip-flip circuits **306-44** and **306-45** in FIG. **8***b*). In some embodiments, filter H(z) may be configured to impose saturation limits (and other types of non-linearities) on one or more state variables. PRN(n) is a sequence of substantially stochastic encoded numerical values, which is combined with (added to) an output from filter H(z). Digital dither sequence x(n) may be a quantized representation of a combination of PRN(n) and an output of filter H(z).

Dither generator **306-41** in FIG. **8***b* is a preferred implementation of mismatch-shaping dither generator **306-40** in FIG. **9**. Specifically, H(z) may be a first-order low-pass filter and have a Z-domain transfer function H(z)=1/(z+1). A numerical value of b(n) may be either 0 or 1. A numerical value of x(n) may be either 0 or 1. A numerical value of PRN(n) may be either +0.5 or −0.5. A pseudo-random sequence PRN(n) may approximate a coin-toss stochastic process. A quantization process may provide a bit value x(n)=1 when an input is greater than 0 and x(n)=0 otherwise. Accordingly, mismatch-shaping dither generator circuit **306-41** may be characterized as a closed-loop control system configured to induce a polarity of a bit-value-difference to suppress a spectral power density in a base-band signal band. Pseudo-random sequence PRN(n) is included to substantially randomize the bit-value-difference sequence, and hence an error induced by mismatch of weighting factors of a CDAC. A calculation of DOUT may include calculating a weighted sum of a plurality of substantially stochastic bit-value-differences, and a bit-value-difference may be characterized by a relatively lower spectral power density in a signal band.

FIGS. **10***a*, **10***b*, **10***c* show a performance of two embodiments of the present teachings compared to a performance of a prior-art SAR ADC. A resolution of a second plurality of capacitors (**201-2** in ADC **200** of FIG. **2** and in ADC **300** of FIG. **6**) may be increased to reduce a quantization error to a very low level (so that properties of a mismatch-induced error signal can be observed more clearly). Specifically the second plurality of capacitors may comprise capacitors with nominal values: C/8, C/8, C/16, C/32, C/64, C/128, C/256, C/512, C/1024, C/1024, C/2048, C/4096, C/8192, C/16384, C/32768, C/65536, C/65536, C/131072, C/262144,

C/524288, C/1048576, C/1048576. It is well-known that voltage-division capacitive structures may be used as a substitute for small capacitors (e.g. C/1048576). It is also well-known that repeat nominal values (here C/8, C/1024, C/65536, C/1048576) may provide over-ranging to suppress potential missing-code type errors. Fully differential ADC structures are used to provide bipolar over-ranging correction. The second plurality of capacitors is coupled to first plurality of capacitors **201-1** (FIG. **2**) in an increased resolution (differential) CDAC configured and operated as CDAC **201** in ADC **200** (FIG. **2**) and in ADC **300** (FIG. **6**). A resolution of digital control circuits **206**, **306** and a number of switch drivers may be increased accordingly.

FIG. **10***a* shows a performance when digital control circuit **206** is configured to operate as in a conventional SAR ADC (i.e., when all bit values in dither code DX and scrambler code SC are set to zero). FIG. **10***b* shows a performance when digital control circuit **206** is configured to operate as described (each sequence of bit values in DX and SC is a substantially random coin-toss sequence). FIG. **10***c* shows a performance when digital control circuit **306** is configured to operate as described (bit values in DX and SC are generated to induce first-order mismatch shaping).

A selected plurality of assumed and actual weighting factors (reflecting mismatch) are the same for each of FIGS. **10***a*, **10***b*, **10***c*. A 4 kHz sinusoidal input signal VIN(t) is provided at a signal level 6 dB below a full scale. A sequence of signal values is sampled periodically at 2048 ksps, and corresponding sequences DOUT(n) are analyzed using a Fourier transformation.

FIG. **10***a* shows an observed spectral power density of DOUT(n) for a conventional SAR ADC. Harmonic distortion is induced by mismatch of weighting factors. A total power of harmonic distortion may be approximately 78 dB below full scale.

FIG. **10***b* shows an observed spectral power density of DOUT(n) for ADC **200** incorporating the increased resolution CDAC. A mismatch-induced error is substantially randomized, and is observed to have a substantially uniform spectral power density (similar to a white-noise signal). A total amount of power of an observed mismatch-induced error may be approximately 73 dB below full scale in a Nyquist range of frequencies (0 Hz to 1024 kHz). A digital filter may be configured to limit a noise bandwidth by a factor of 8 to 128 kHz, and to provide a digital signal having an increased accuracy. Specifically, a total amount of power of a mismatch-induced error comprised in a frequency range from 0 Hz to 128 kHz may be approximately 82 dB below full scale. If a noise bandwidth is instead reduced by a factor of 16 or 32, an inband power of a mismatch-induced error may be reduced to approximately 85 dB or 88 dB below full scale.

A digital filter may be configured to reduce a noise bandwidth of DOUT(n) provided by a conventional SAR ADC. However, substantial harmonic distortion may be present in a signal band. For example, a power of harmonic distortion shown in FIG. **10***a* may be reduced by less than 1 dB when a digital filter is configured to reduce a bandwidth by a factor of 32. Accordingly, a digital filter may be configured to perform an averaging operation whereby an accuracy of ADC **200** may be improved substantially more than an accuracy of a prior-art SAR ADC may be improved by the same digital filter.

FIG. **10***c* shows an observed spectral power density of DOUT(n) for ADC **300** (FIG. **6**) incorporating the increased resolution CDAC. A mismatch-induced error is substantially randomized, and may be observed to have a relatively lower spectral power density at low frequencies. A total amount of power of an observed mismatch-induced error may be approximately 73 dB below full scale in a Nyquist range of frequencies (0 Hz to 1024 kHz). A digital filter may be configured to limit a noise bandwidth by a factor of 8, 16, or 32 and provide a digital signal wherein an inband power of a mismatch-induced error may be approximately 91 dB, 100 dB, or 109 dB below full scale. Accordingly, ADC **300** may be configured to provide a greater bandwidth and/or a better accuracy than ADC **200** or a prior-art SAR ADC.

FIG. **11***a* shows a third embodiment of the present teachings ADC **400**. ADC **400** comprises an ADC circuit **300** that may be identical to (or a variation of) ADC **300** of FIG. **6**. ADC **400** further comprises a switched-capacitor circuit comprising an amplifier circuit **401**, a capacitor **402**, and a plurality of switches **403**, **404**, **405**. At an end of an A/D conversion operation of ADC **300**, switch **403** is opened, and switch **404** is closed (switch **405** is already closed), whereby amplifier **401** is configured in a negative feedback configuration with capacitor **402** providing negative feedback. A charge portion representing a residue at sampling node **204** is transferred to capacitor **402** by amplifier **401**. After a settling period of time, sufficiently long to allow for a substantially complete transfer of the residue, switch **405** is opened to sample the residue on capacitor **402**. After sampling the residue on capacitor **402**, ADC **300** is configured in a sampling configuration to acquire a next value VIN of VIN(t) in a manner described for ADC **300** of FIG. **6**. Sampling switch **208** is opened at a sampling instance. A substantially isolated charge portion on sampling node **204** is a sampled value representing a combination of VIN and an analog dither value Vdither(DX) for a digital dither code DX provided by ADC **300**. In sequence after the sampling instance, switch **405** is closed, switch **404** is opened, and switch **403** is closed. A charge portion representing a residue from a previous A/D conversion operation, sampled on capacitor **402** as described, is thereby transferred back to sampling node **204**. Accordingly, for the A/D conversion operation, sampling node **204** represents an analog value comprising a portion of an analog signal value, a portion of an analog dither value, and a portion of residue sampled in a preceding A/D conversion operation. ADC **300** is then configured to derive a code DY, and a next residue is transferred to capacitor **402** by again configuring switches **403**, **404**, **405** as described. ADC **300** is configured to combine digital codes DX and DY to derive an encoded numerical representation DOUT of VIN.

FIG. **11***b* shows a signal-flow model for ADC **400**. VIN(n)+Vdither(DX(n)) may represent a sequence of sampled values substantially isolated on sampling node **204** at sampling instances. U(n) may represent residues sampled on capacitor **402** and transferred to sampling node **204** after sampling instances. Q(n) may represent charge portions on sampling node **204** corresponding to codes DY(n) derived by ADC **300**. E(n) may be a sequence of residue charge portions left on sampling node **204** after codes DY(n) are derived by ADC **300** and then transferred to capacitor **402**. A filter with transfer function 1-NTF(z) may represent a delaying operation of sampling residues E(n) on capacitor **402** and adding them back one conversion cycle later, U(n)=E(n−1). Accordingly, using a Z-transformation, 1−NTF(z)=1/z, implying that a noise transfer function may be NTF(z)=1−1/z.

Those skilled in the art will recognize that the signal-flow model shown in FIG. **11***b* is that of a delta-sigma modulator configured in a so-called error-feed-back (EFB) topology. Accordingly, ADC **400** of FIG. **11***a* may be configured to randomize and shape a quantization error of DOUT according to a noise transfer function NTF(z)=1−1/z. ADC **400** may be configured to provide first-order suppression of quantiza-

tion errors in a signal band. ADC **400** may further be configured to suppress mismatch-induced errors in the signal-band. ADC **400** is configured to combine a sampled value with a residue sampled in a preceding analog-to-digital conversion operation.

Several variations of the third embodiment are envisioned. In another embodiment, residues from several previous A/D conversion operations may be represented in a switched-capacitor circuit and they may be combined with a sampled value to achieve a higher degree of randomization and/or shaping of quantization errors. For example, an operation may be described by signal-flow model FIG. **11***b*, and a noise transfer function NTF(z) may be a second, third, or fourth order differentiation, a notch filter, or any other noise transfer function for use with a delta-sigma modulator.

Comparator circuit **205** may include one or more amplifier circuits. In another embodiment, amplifier circuit **401** may be implemented by circuitry that is included into comparator circuit **205**. In another embodiment, amplifier circuit **401** may be a distinct (separate) circuit, and comparator circuit **205** may be coupled to an output of amplifier **401**. In that case, switch **403** may be removed (always open), and switch **404** may be substituted by a wire (always closed). In another embodiment, amplifier circuit **401** may be used to sample a residue on capacitor **402** at an end of a first A/D conversion operation, and be configured and reused as an amplifier **218** in FIG. 4 of US Patent Application Publication US2011/0115661 A1 during a subsequent second A/D conversion operation.

In another embodiment, a CDAC may be configured to have a resolution exceeding a resolution of a digital code DY derived by successive approximation (or otherwise). Several capacitors in the CDAC may be configured to apply an analog dither value Vdither(DX) during an acquisition period, and need not be applied a bit value of DY during an A/D conversion operation. A digital dither code DX applied to a CDAC may thus have a least-significant-bit step size that is smaller than a least-significant-bit step size of an A/D conversion process. Some bit values in a digital dither code DX may be substantially random, while other bit values in DX may be generated to shape a spectral power density of a bit-value-difference sequence. Some input terminals of a CDAC may be applied a pre-determined fixed value during acquisition periods.

FIG. **12***a* shows a fourth embodiment of the present teachings ADC **500**. FIG. **12***b* shows a timing diagram for ADC **500**, illustrating when a plurality of switches (labeled in FIG. **12***a* by timing control signals SAMP, PH**1**, PH**2**, and PH**3**) are open and closed. FIG. **12***c* shows a signal-flow model of ADC **500**. ADC **500** is similar to ADC **400** in that it is configured to substantially randomize and shape a quantization error of an ADC circuit **300**. ADC circuit **300** may be identical to (or a variation of) ADC **300** of FIG. **6**. After an A/D conversion operation, when timing control signal PH**2** is high (FIG. **12***b*), a switched-capacitor circuit **511** (FIG. **12***a*) is configured to transfer a charge portion representing a residue R(n) (FIG. **12***c*) from sampling node **204** (FIG. **12***a*) to a capacitor **501**. Capacitor **501** and an amplifier **502** are configured to provide a voltage that is substantially a running sum (integration) of residues R(n) with a negative polarity. A capacitor **503** is switched to transfer charge portions to another capacitor **504** and an amplifier **505** configured to provide a voltage that substantially represents a running sum of the running sum (second-order integration) of R(n) with a negative polarity. At a beginning of an A/D conversion operation, amplifiers **502**, **505** provide voltages that represent a history (a filtered version) of residues R(n) for a plurality of previous A/D conversion operations. Accordingly, residues R(n) are sampled at an output **204** of CDAC **201** and provided as an input to switched capacitor filter **511**. Switched-capacitor filter **511** combines a value sampled on node **204** with residues from a first preceding ADC operation, a second preceding ADC operation, a third preceding ADC operation, et cetera.

ADC **300** (FIG. **12***a*) samples an input voltage signal VIN (t) in a manner described for ADC **300** of FIG. **6**. A substantially isolated charge portion on sampling node **204** is a sampled value representing a combination of a signal value VIN and an analog dither value Vdither(DX). During acquisition periods, a capacitor **506** and a capacitor **507** are coupled to predefined potentials (ground). After a sampling instance, at a beginning of an A/D conversion operation, capacitors **506**, **507** are switched to outputs of amplifiers **502**, **505** to add a filtered version T(n) of R(n) to the sampled value on sampling node **204**. Accordingly, for the A/D conversion operation, sampling node **204** represents an analog value comprising a portion of an analog signal value, a portion of an analog dither value, and portions for a plurality of residues sampled in preceding A/D conversion operations. Polarity-inverting circuits **508**, **509** may be implemented by interchanging a positive and a negative output terminal for each amplifier **502**, **505** in a fully-differential implementation of ADC **500** and switched-capacitor circuit **511** (well-known by someone ordinarily skilled in the art). Accordingly, as shown in FIG. **12***c*, a filter H(z) receives a sampled residue signal R(n) and provides a filtered version T(n) of R(n). T(n) is added to a sampled value VIN(n)+Vdither(DX(n)) and provided as an input to an ADC. An impulse response of transfer function H(z) may characterize T(n) as a weighted sum of residues from preceding A/D conversion operations.

ADC **300** is configured to derive a code DY for which a residue on sampling node **204** is within a predefined range corresponding to a resolution (quantization). Accordingly, DY(n) may be a quantized representation of VIN(n)+Vdither(DX(n))+T(n) as shown in FIG. **12***c*. Notice that capacitors **506**, **507** are switched back to the predefined potential (ground) when switch **510** is closed by timing control signal PH**2**. Accordingly, residue R(n) transferred to capacitor **501** is a residue of the sampled value VIN+Vdither(DX) with respect to the derived code DY. The filtered version T(n) is only temporarily transferred to (represented on) sampling node **204** during the A/D conversion operation to derive digital code DY; it is subsequently removed and substantially not included in residue R(n) transferred to capacitor **501**. In another embodiment, other means may be used to provide a signal path from T(n) to DY(n). For example, capacitors **506**, **507** (FIG. **12***a*) may be removed, and circuits **508**, **509** may be substituted by transconductance stages providing currents configured to induce an offset in comparator circuit **205**. A delta-sigma circuit is provided for randomizing and shaping a quantization error when a circuit is configured for a weighted sum (a filtered version) of residues from previous A/D conversion operations to induce a bias in a quantization error of a encoded numerical representation DOUT of an analog signal value.

It will be understood that individual signal paths shown in FIG. **12***c* do not necessarily correspond to distinct physical circuit nodes of ADC **500**. Those who are skilled in the art will recognize that the signal-flow model shown in FIG. **12***c* is that of a delta-sigma modulator, and that a transfer function H(z) of switched-capacitor filter **511** may be selected to obtain a desired noise transfer function for substantially randomizing and shaping a quantization error. For example, H(z) may be a

cascade of any number of integrators configured with local feedback paths for resonance to provide a large amount of gain in a selected signal band.

A fifth embodiment of the present teachings may have a structure similar to ADC **200** of FIG. **2**. Digital control circuit **206** (FIG. **3**) may be modified to implement an A/D conversion operation wherein all values of scrambler code SC are used during the A/D conversion operation of a single value sampled on CDAC **201**. Specifically, a first successive-approximation operation may provide a first digital code DY_00 for SC='00'. A second successive-approximation operation may provide a second digital code DY_01 for SC='01'. A third successive-approximation operation may provide a third digital code DY_10 for SC='10'. A fourth successive-approximation operation may provide a fourth digital code DY_11 for SC='**11**'. An encoded numerical representation DOUT may be an average value of DOUT_00 (calculated as described herein for DX, and DY=DY_00), DOUT_01 (calculated as described herein for DX, and DY=DY_01), DOUT_10 (calculated as described herein for DX, and DY=DY_10), and DOUT_11 (calculated as described herein for DX, and DY=DY_11). A CDAC for use with this fifth embodiment may preferably incorporate over-ranging, so that digital codes DY_01, DY_10, and DY_11 can be derived with relatively few steps of successive-approximation once code DY_00 is determined. Differences between DY_00, DY_01, DY_10, and/or DY_11 may reflect mismatch of weighting factors w(T1), w(T2), w(T3), and w(T4), and a CDAC may be configured to incorporate an over-range that can accommodate such differences without changing a first plurality of bit values (e.g. b1, b2, b3, b4, b5, b6, b7, b8). An aspect of the fifth embodiment is that bit values b1, b2, b3, b4 scrambled by scrambler circuit **206-6** are all substantially represented by one common weighting factor w(T1234)=w(T1)/4+w(T2)/4+w(T3)/4+w(T4)/4. Accordingly, mismatch of weighting factors w(T1), w(T2), w(T3), w(T4) are suppressed by an averaging operation that is internal to a single A/D conversion operation (unlike an averaging operation implemented by a digital filter, which may be viewed as averaging across several A/D conversion operations).

In other embodiments of the present teachings, an averaging operation of a plurality of weighting factors that is internal to a single A/D conversion operation (similar to the described fifth embodiment) may be combined with ADC **300** of FIG. **6**, ADC **400** of FIG. **11***a*, and/or ADC **500** of FIG. **12***a*.

The present teachings have described several embodiments incorporating a digital control circuit for successive approximation. Those skilled in the art will recognize that the present teachings are not dependent on how a digital code is derived, and that a successive-approximation A/D conversion operation is provided only as an example. Other embodiments of the present teachings may use other circuits and/or methods to provide a digital code representing a value sampled on a CDAC. For example, US Patent Application Publication US 2011/0115661 A1 describes circuits and methods for deriving a digital code in relatively fewer steps (e.g. a 16-bit code may be derived in 4 steps). US Patent Application Publication US 2011/0115661 A1 further describes how a CDAC may be used to provide negative feedback for an amplifier circuit, and that a multi-bit comparator circuit (flash ADC) may be configured to evaluate a residue at a node of a CDAC other than at a sampling node. The present teachings may advantageously be combined with the teachings of US Patent Application Publication US 2011/0115661 A1 (and many other types of analog-to-digital converters).

A binary-to-thermometer encoder circuit, or another dedicated encoder circuit, may be used to bring a digital code from a SAR (or other) digital control circuit to a format that is required for a particular type of scrambler circuit for a mismatch-shaping encoder. An input to scrambler **206-6** of FIGS. **4***a* and **4***b* (and other known scrambler circuits) need not be encoded in a thermometer-code format when it is operated as a zero-order mismatch-shaping encoder as in ADC **200** (i.e., controlled by a substantially random scrambler code SC). For example, digital state machine **206-1** (FIG. **3**) may be configured to skip a step described herein to determine bit value b2, and instead set bit value b2 equal to bit value b1 (thus shortening a conversion time).

Various types of scrambler circuits may be provided to implement a mismatch-shaping encoder. Well known examples include a rotary scrambler (U.S. Pat. No. 5,221,926), a butterfly scrambler (U.S. Pat. No. 5,404,142), a tree scrambler (U.S. Pat. No. 5,684,482), and others. A scrambler circuit may be controlled by a scrambler code selecting a permutation supported by the scrambler circuit in response to one or more random bit values and/or bit values provided as an input to the scrambler circuit. Some scrambler circuits support more permutations than other scrambler circuits do, and a resolution of a scrambler code may vary from one type of scrambler circuit to another type. A permutation of N terminals may be described by a sequence (an ordered plurality of input terminals being re-ordered into a sequence for an ordered plurality of output terminals). If a digital code is determined sequentially in a plurality of steps, and a first and a second bit value is determined in any one step, then no distinction shall be made with respect to whether the first bit value is determined before or after the second bit value. A scrambler circuit may provide two-or-more permutations for two-or-more bit values. Some scrambler circuits (e.g. tree-type scrambler circuits) may include an encoding function (e.g., binary-weighted to unit-weighted codes), and may not have the same number of input and output terminals. A permutation/sequence for such scrambler circuits may be determined/defined after separating the composite function into a separate encoding function and a separate scrambling function.

A scrambler circuit need not be a circuit block configured between a digital state machine and a CDAC. A comparator circuit may comprise a scrambler circuit and provide a scrambled multi-bit indication of an output of a CDAC. The scrambled multi-bit indication may be applied to the CDAC without further scrambling. For example, a flash ADC may be a comparator circuit providing a scrambled multi-bit indication when a scrambler circuit is configured to provide a scrambled plurality of reference potentials (instead of a conventional fixed plurality of reference potentials, e.g., provided by a resistive reference ladder).

Some embodiments of the present teachings do not require a scrambler circuit. For example, a variation of ADC **300** may be configured to operate with a CDAC having only one capacitor (C/4) instead of first plurality of capacitors **201-1**. VIN(t) may be sampled on the one capacitor (C/4) and all other capacitors in the CDAC may be used to apply a digital dither code DX. Only one permutation of a single bit value exists (all permutations are realized by a wire substituting scrambler circuit **206-6**). Accordingly, embodiments of the present teachings may be configured to operate without a scrambler circuit.

Some embodiments of the present teachings may use one or more separate structures to sample one or more signal values at a sampling instance. A sampled signal value may subsequently (i.e., after the sampling instance) be transferred to a CDAC for an A/D conversion operation. A signal value sampled on a capacitor may be transferred to a CDAC by

connecting a terminal of the capacitor to a sampling node of the CDAC. In some embodiments, a differential analog voltage signal value may be applied to a first and a second input terminal of a capacitive structure at a sampling instance, and the first and the second input terminals may be shorted during an analog-to-digital conversion operation (and not necessarily applied a reference voltage potential).

Some embodiments of the present teachings may be configured to operate with assumed weighting factors determined at design time. Other embodiments of the present teachings may combine bit values of codes applied to a CDAC circuit with codes representing estimated values of one or more assumed weighting factors.

In some embodiments, a mismatch-induced error may be relatively small and may be substantially masked by other imperfections such as thermal noise. Accordingly, some embodiments of the present teachings may be configured to operate as Nyquist-rate analog-to-digital converters (not involving any averaging by a digital filter).

Numerous variations of the present teachings are envisioned. Embodiments described herein are merely illustrations of exemplary embodiments. An appropriate choice of circuitry may depend on a specific application and other factors such as available types of semiconductors, capacitors, resistors, reliability voltage limits, silicon area, cost, and additional factors and considerations typically involved in a design of an integrated circuit. For example, a digital control circuit may be implemented as a state machine in a CMOS technology, or using any other known circuit technique, method and/or process technology suitable for an implementation of such circuits. Various embodiments may incorporate switches implemented as CMOS transmission-gate switches, bootstrapped switches, single-device switches, and/or any other suitable switching devices. For example, an operation of a switch shall include circuits of a type known as "switched-opamp", wherein a switch is an implicit aspect of controlling an output impedance of an amplifier. An ADC system implemented according to the present teachings may incorporate a plurality of types of semiconductor devices (including all flavors of MOS, BJT, IGBT, IGFET, JFET, FINFET, organic transistors, nano-carbon-tube devices, electro-mechanical switches etc.), some of which may be selected to withstand a high-voltage input signal VIN(t), and some of which may be selected for fast settling of low-voltage circuit nodes. Accordingly, an ADC system may be implemented using a process technology that provides asymmetrical devices (BCD etc.) in addition to symmetrical MOS devices, and a process technology may incorporate oxides and other physical structures having a plurality of dimensions and electrical properties. Those who are skilled in the art will recognize that ADC systems **200**, **300**, **400**, and **500** (depicted in FIGS. **2**, **6**, **11***a*, **12***a*) and any other single-ended embodiment of the present teachings may be modified to a similar fully-differential exemplary embodiment of the present teachings. Accordingly, an ADC system according to the present teachings may interface differential signals with an arbitrary common-mode voltage level. Other embodiments may interface single-ended signals (unipolar or bipolar) defined with respect to a reference voltage potential that may or may not be sampled substantially at the same time as an input voltage signal is sampled (pseudo-differential operation). A full-scale range of an ADC system according to the present teachings may be substantially a reference voltage range. Alternatively, an ADC system according to the present teachings may incorporate analog scaling and/or digital scaling techniques to provide one or more selectable full-scale ranges that are either wider or narrower than a reference voltage range. Input signals may be provided by active and/or passive circuits, which may be designed to optimize various performance parameters, such as settling time, noise bandwidth, distortion, power consumption, etc. An ADC system according to the present teachings may consume very little power (including substantially zero power) during an acquisition period when a capacitor is configured to be charged according to an input signal while active circuitry is in a power-down (or other reduced-power) configuration. A common-mode shift, charge-pumping, and other techniques may be applied during one or more steps of an A/D conversion process to make circuitry operate with respect to a desired operating point (e.g. to substantially isolate a charge portion and/or to avoid substantial forward-biasing of isolating PN junctions, or to provide a desired frequency response, gain function, etc.). A comparator circuit may provide an indication of a polarity, an indication of a potential (e.g., flash ADC), or some other type of indication that may be used to derive or refine a code representing a sampled value. For example, an oscillator controlled by a residue (e.g., a VCO) may provide an indication as a phase shift that may occur during a period of time. Redundancy and over-ranging may be incorporated in various embodiments of the present teaching. A degree of redundancy and over-ranging may be selected to meet specific objectives, such as conversion rate, latency, resolution, environmental parameters, circuit complexity, etc.

Various types of pseudo-random-number (PRN) generators are known to someone ordinarily skilled in the art. For example, a PRN generator may be implemented as a LFSR circuit and/or a CASR circuit providing one or more sequences of bit values that may repeat after a period of time (split seconds, minutes, or thousands of years). Improved pseudo-stochastic properties may be achieved by combining multiple different PRN structures. For example, a first-length sequence of bit-values from a LFSR circuit may be combined with a second-length sequence from a CASR circuit by an exclusive-or operation. Further improved stochastic properties may be obtained by combining a PRN bit-value sequence with a bit-value sequence obtained by quantizing a physical-noise process (e.g., thermal noise). Non-conventional structures may be used to generate bit-value sequences that may be classified as substantially random. For example, PRN5 in mismatch-shaping dither generator **306-4** (FIG. **8***a*) may be provided by applying a non-linear operation to a plurality of bit values (e.g., x6 EXOR x7), preferably including a bit value obtained from a physical stochastic process (e.g., a least-significant bit from an A/D conversion operation that is substantially limited by thermal noise).

Any known method to overcome or suppress imperfections of an analog-to-digital converter circuit (or sub-circuit, e.g. a reference voltage circuit) is envisioned to be used in combination with this present teaching. This present teaching may be incorporated as a subsystem in a larger ADC system (e.g., it may be combined with other types of ADCs, including pipeline ADCs, SAR ADCs, delta-sigma ADCs, etc.). The present teachings may be embodied in electrical and/or electro-mechanical systems of a higher degree of functional complexity, such as industrial control systems, medical applications (e.g., x-ray and MRI machines), consumer application (e.g., games and television), et cetera. An ADC system according to the present teachings may provide multiple channels to interface several distinct analog signals, e.g., via a multiplexing front-end circuit and/or an array of sample-and-hold circuits. A mismatch-shaping dither generator (e.g. **306-41** in FIG. **8***b*) may operate on a per-channel basis, such that registers (e.g., flip-flop circuits **306-44**, **306-45** in FIG. **8***b*) are implemented for each input channel and updated only

when a sample from a particular input channel is A/D converted. An ADC system implemented according to the present teachings may be implemented on a single semiconductor substrate, or as multiple semiconductors in a package, or as several devices assembled on a printed circuit board (or otherwise). An ADC implemented according to the present teaching may incorporate timing circuits causing transitions between various configurations, and/or it may receive timing signals provided by an external source. Digital codes and numerical representations of analog signal values may be encoded in various formats, and may be communicated to external systems in a serial format, a parallel format, or some other format deemed suitable or advantageous for an application. Such digital codes may be communicated (and power may be provided) via an interface providing galvanic isolation to limit interference, enhance safety, or to derive some other benefit.

Accordingly, while particular embodiments of the present teaching have been shown and described, it will be obvious to those skilled in the art that changes and modifications may be made without departing from this present teaching in its broader aspects and, therefore, the appended claims are to encompass within their scope all such changes and modifications that are within the true spirit and scope of this present teaching.

What is claimed is:

1. An analog-to-digital converter system comprising:

a capacitive digital-to-analog converter configured to sample a combination of an analog signal value and an analog dither value; and

a control circuit comprising a scrambler circuit for selectively permutating bit values of digital codes applied to the capacitive digital-to-analog converter; the control circuit configured to sequentially apply a plurality of digital codes to the capacitive digital-to-analog converter during an analog-to-digital conversion operation to derive an encoded numerical representation of the combination of the analog signal value and the analog dither value.

2. The analog-to-digital converter system of claim **1**, wherein the scrambler circuit is configured for dynamic element matching.

3. The analog-to-digital converter system of claim **1**, wherein the scrambler circuit is configured to operate as a scrambler for a mismatch-shaping encoder.

4. The analog-to-digital converter system of claim **1**, wherein the scrambler circuit is configured to suppress in a signal band an error induced by mismatch of weighting factors of a plurality of input terminals of the capacitive digital-to-analog converter.

5. The analog-to-digital converter system of claim **1**, wherein the scrambler circuit is configured to substantially randomize an error signal induced by mismatch of weighting factors of a first plurality of input terminals of the capacitive digital-to-analog converter.

6. The analog-to-digital converter system of claim **5**, further comprising a dither generator circuit configured to provide the analog dither value; the dither generator circuit configured to substantially randomize an error signal induced by mismatch of weighting factors of a second plurality of input terminals of the capacitive digital-to-analog converter.

7. The analog-to-digital converter system of claim **1**, wherein a calculation of an encoded numerical representation of the analog signal value includes calculating a weighted sum of a plurality of substantially stochastic bit-value-differences.

8. The analog-to-digital converter system of claim **7**, wherein a substantially stochastic bit-value-difference in the plurality of substantially stochastic bit-value-differences is characterized by a relatively lower spectral power density in a signal band.

9. The analog-to-digital converter system of claim **1**, further comprising a dither generator circuit configured to apply a digital dither code to the capacitive digital-to-analog converter at a sampling instance.

10. The analog-to-digital converter system of claim **9**, wherein the control circuit is configured to combine the digital dither code with a digital code derived to represent the combination of the analog signal value and the analog dither value to derive an encoded numerical representation of the analog signal value.

11. The analog-to-digital converter system of claim **9**, wherein a bit value of the digital dither code is a substantially random quantity.

12. The analog-to-digital converter system of claim **9**, wherein the dither generator circuit is configured to substantially randomize and induce stochastic properties for a plurality of bit values in a digital code derived by the control circuit to represent the combination of the analog signal value and the analog dither value.

13. The analog-to-digital converter system of claim **9**, wherein the dither generator circuit is configured to be responsive to a digital code derived by the control circuit in a preceding analog-to-digital conversion operation to represent a sampled value.

14. The analog-to-digital converter system of claim **9**, wherein the dither generator circuit in a first mode of operation is configured to set a bit value of the digital dither code responsive to a bit value of a digital code derived by the control circuit in a preceding analog-to-digital conversion operation to represent a sampled value.

15. The analog-to-digital converter system of claim **14**, wherein the dither generator circuit is configured to switch between the first mode of operation and a second mode of operation, the dither generator circuit in the second mode of operation configured to set said bit value of said digital dither code on a substantially random basis.

16. The analog-to-digital converter system of claim **1**, including circuitry configured to store digital codes for weighting factors of the capacitive digital-to-analog converter estimated after the analog-to-digital converter system is manufactured.

17. The analog-to-digital converter system of claim **1**, including circuitry configured to perform at least one step of successive approximation during the analog-to-digital conversion operation.

18. The analog-to-digital converter system of claim **1**, wherein the control circuit is configured to provide an indication of a residue of the combination of the analog signal value and the analog dither value with respect to a digital code applied to the capacitive digital-to-analog converter.

19. The analog-to-digital converter system of claim **1**, wherein the scrambler circuit is configured to apply at least two distinct permutations during the analog-to-digital conversion operation.

20. The analog-to-digital converter system of claim **1**, wherein the control circuit is configured to derive a first digital code representing the combination of the analog signal value and the analog dither value when the scrambler circuit is configured to apply a first permutation; the control circuit is further configured to derive a second digital code representing the combination of the analog signal value and the analog dither value when the scrambler circuit is configured to apply

a second permutation distinct from the first permutation; the control circuit is further configured to combine the first digital code, the second digital code, and a digital dither code to derive an encoded numerical representation of the analog signal value.

21. The analog-to-digital converter system of claim **1**, further comprising a digital filter configured to limit a noise bandwidth.

22. The analog-to-digital converter system of claim **1**, including circuitry configured to apply an averaging operation.

23. The analog-to-digital converter system of claim **1**, wherein the control circuit is configured to substantially randomize and suppress in a signal band a spectral power density of a quantization error.

24. The analog-to-digital converter system of claim **1**, further comprising a switched-capacitor filter configured to receive a residue signal comprising a quantization error.

25. The analog-to-digital converter system of claim **24**, wherein the switched-capacitor filter is configured to randomize and suppress the quantization error in a signal band.

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